

Ustekinumab

Policy History

- Last Review 06/23/2025

Effective: 11/22/2016

Next Review: 03/26/2026

- Review History
- Definitions

Additional Information

- Clinical Policy Bulletin Notes

Number: 0912

Table Of Contents

Policy
Applicable CPT / HCPCS / ICD-10 Codes
Background
References

Brand Selection for Medically Necessary Indications for Commercial Medical Plans

Note: For plaque psoriasis, this policy applies to all members (new starts and continuation of therapy) requesting treatment with a targeted immune modulator. For all other indications, this policy applies to members who are new to treatment with a targeted immune modulator for the first time.

As defined in Aetna commercial policies, health care services are not medically necessary when they are more costly than alternative services that are at least as likely to produce equivalent therapeutic or diagnostic results. Imuldosa (ustekinumab-srlf), Otulfi and unbranded Otulfi (ustekinumab-aauz), Pyzchiva and unbranded Pyzchiva (ustekinumab-ttwe), Selarsdi and unbranded Selarsdi (ustekinumab-aekn), Starjemza (ustekinumab-hmny), Steqeyma and unbranded Steqeyma (ustekinumab-stba), unbranded Stelara (ustekinumab), Wezlana (ustekinumab-auub), and Yesintek (ustekinumab-kfce) are more costly to Aetna than other Ustekinumab products. There is a lack of reliable evidence that Imuldosa (ustekinumab-srlf), Otulfi and unbranded Otulfi (ustekinumab-aauz), Pyzchiva and unbranded Pyzchiva (ustekinumab-ttwe), Selarsdi and unbranded Selarsdi (ustekinumab-aekn), Starjemza (ustekinumab-hmny), Steqeyma and unbranded Steqeyma (ustekinumab-stba), unbranded Stelara (ustekinumab), Wezlana (ustekinumab-auub), and Yesintek (ustekinumab-kfce) are superior to the lower cost Ustekinumab product: Stelara (ustekinumab). Therefore, Aetna considers Imuldosa (ustekinumab-srlf), Otulfi and unbranded Otulfi (ustekinumab-aauz), Pyzchiva and unbranded Pyzchiva (ustekinumab-ttwe), Selarsdi and unbranded Selarsdi (ustekinumab-aekn), Starjemza (ustekinumab-hmny), Steqeyma and unbranded Steqeyma (ustekinumab-stba), unbranded Stelara (ustekinumab), Wezlana (ustekinumab-auub), and Yesintek (ustekinumab-kfce) to be medically necessary only for members who have a contraindication, intolerance or ineffective response to the available equivalent alternative targeted immune modulators per criteria below:

Moderately to Severely Active Crohn's Disease (CD)

For the treatment of CD, member has a contraindication, intolerance or ineffective response to all of the following available equivalent alternative targeted immune modulators (one-month trial each): both Entyvio, Stelara and either Avsola, Inflectra, or Renflexis.

Moderately to Severely Active Ulcerative Colitis (UC)

For the treatment of UC, member has a contraindication, intolerance or ineffective response to all of the following available equivalent alternative targeted immune modulators (one-month trial each): both Entyvio, Stelara and either Avsola, Inflectra or Renflexis.

Active Psoriatic Arthritis (PsA)

For the treatment of PsA, member has a contraindication, intolerance or ineffective response to all of the following available equivalent alternative targeted immune modulators (one-month trial each): both Simponi Aria, Stelara and either Avsola, Inflectra, or Renflexis.

Moderate to Severe Plaque Psoriasis

For the treatment of chronic severe plaque psoriasis, member has a contraindication, intolerance or ineffective response to all of the following available equivalent alternative targeted immune modulators (one-month trial): both Ilumya, Stelara, and either Avsola, Inflectra, or Renflexis.

Policy

Scope of Policy

This Clinical Policy Bulletin addresses the following ustekinumab products for commercial medical plans:

- ustekinumab (Stelara and unbranded Stelara)
- ustekinumab-aaaz (Otulfi and unbranded Otulfi)
- ustekinumab-aekn (Selarsdi and unbranded Selarsdi)
- ustekinumab-auub (Wezlana)
- ustekinumab-hmny (Starjemza)
- ustekinumab-kfce (Yesintek)
- ustekinumab-srlf (Imuldosa)
- ustekinumab-stba (Steqeyma and unbranded Steqeyma)
- ustekinumab-ttwe (Pyzchiva and unbranded Pyzchiva).

For Medicare criteria, see Medicare Part B Criteria.

Note: Requires Precertification:

Precertification of ustekinumab products is required of all Aetna participating providers and members in applicable plan designs. For precertification of ustekinumab products, call (866) 752-7021 or fax (888) 267-3277. For Statement of Medical Necessity (SMN) precertification forms, see Specialty Pharmacy Precertification.

1. Prescriber Specialties

This medication must be prescribed by or in consultation with one of the following:

1. Plaque Psoriasis: dermatologist;
2. Psoriatic arthritis: rheumatologist or dermatologist;
3. Crohn's disease and ulcerative colitis: gastroenterologist;
4. Immune checkpoint inhibitor-related toxicity: gastroenterologist, hematologist or oncologist.

2. Criteria for Initial Approval

Aetna considers ustekinumab, or its biosimilar, medically necessary for the following indications when criteria are met:

1. Plaque psoriasis (PsO)

1. For members 6 years of age or older who have previously received a biologic or targeted synthetic drug (e.g., Sotyktu, Otezla) indicated for the treatment of moderate to severe plaque psoriasis; *or*
2. For members 6 years of age or older for the treatment of moderate to severe plaque psoriasis when *any* of the following criteria is met:

1. Crucial body areas (e.g., hands, feet, face, neck, scalp, genitals/groin, intertriginous areas) are affected; *or*
2. At least 10% of the body surface area (BSA) is affected; *or*
3. At least 3% of body surface area (BSA) is affected and the member meets *either* of the following criteria:
 1. Member has had an inadequate response or intolerance to either phototherapy (e.g., UVB, PUVA) or pharmacologic treatment with methotrexate, cyclosporine, or acitretin; *or*
 2. Member has a clinical reason to avoid pharmacologic treatment with methotrexate, cyclosporine and acitretin (see Appendix);

2. Psoriatic arthritis (PsA)

1. For members 6 years of age or older who have previously received a biologic or targeted synthetic drug (e.g., Rinvoq, Otezla) indicated for active psoriatic arthritis; *or*
2. For members 6 years of age or older for treatment of active psoriatic arthritis when *either* of the following criteria is met:
 1. Member has mild to moderate disease and meets *one* of the following criteria:
 1. Member has had an inadequate response to methotrexate, leflunomide, or another conventional synthetic drug (e.g., sulfasalazine) administered at an adequate dose and duration; *or*
 2. Member has an intolerance or contraindication to methotrexate or leflunomide (see Appendix), or another conventional synthetic drug (e.g., sulfasalazine); *or*
 3. Member has enthesitis; *or*
 2. Member has severe disease;

3. Crohn's disease (CD)

For treatment of moderately to severely active CD;

4. Ulcerative colitis (UC)

For treatment of moderately to severely active UC;

5. Immune checkpoint inhibitor-related toxicity

For the treatment of immune checkpoint inhibitor-related diarrhea or colitis when the member has experienced an inadequate response, intolerance, or contraindication to infliximab or vedolizumab.

Aetna considers all other indications as experimental, investigational, or unproven.

3. Continuation of Therapy

Aetna considers continuation of ustekinumab, or its biosimilar, medically necessary for the following indications:

1. Plaque psoriasis (PsO)

For all members 6 years or older (including new members) who are using the requested medication for moderate to severe plaque psoriasis and who achieve or maintain a positive clinical response as evidenced by low disease activity or improvement in signs and symptoms of the condition when *either* of the following is met:

1. Reduction in body surface area (BSA) affected from baseline; *or*
2. Improvement in signs and symptoms from baseline (e.g., itching, redness, flaking, scaling, burning, cracking, pain);

2. Psoriatic arthritis (PsA)

For all members 6 years of age or older (including new members) who are using the requested medication for psoriatic arthritis and who achieve or maintain positive clinical response as evidenced by low disease activity or improvement in signs and symptoms of the condition when there is improvement in *any* of the following from baseline:

1. Number of swollen joints; *or*
2. Number of tender joints; *or*
3. Dactylitis; *or*
4. Enthesitis; *or*
5. Skin and/or nail involvement; *or*
6. Functional status; *or*
7. C-reactive protein (CRP);

3. Crohn's Disease (CD)

1. For all members (including new members) who are using the requested medication for moderately to severely active Crohn's disease and who achieve or maintain remission; *or*
2. For all members (including new members) who are using the requested medication for moderately to severely active Crohn's disease and who achieve or maintain a positive clinical response as evidenced by low disease activity or improvement in signs and symptoms of the condition when there is improvement in *any* of the following from baseline:
 1. Abdominal pain or tenderness; *or*
 2. Diarrhea; *or*
 3. Body weight; *or*
 4. Abdominal mass; *or*
 5. Hematocrit; *or*
 6. Appearance of the mucosa on endoscopy, computed tomography enterography (CTE), magnetic resonance enterography (MRE), or intestinal ultrasound; *or*
 7. Improvement on a disease activity scoring tool (e.g., Crohn's Disease Activity Index [CDAI] score);

4. Ulcerative colitis

1. For all members (including new members) who are using the requested medication for moderately to severely active ulcerative colitis and who achieve or maintain remission; *or*
2. For all members (including new members) who are using the requested medication for moderately to severely active ulcerative colitis and who achieve or maintain a positive clinical response as evidenced by low disease activity or improvement in signs and symptoms of the condition when there is improvement in *any* of the following from baseline:
 1. Stool frequency; *or*
 2. Rectal bleeding; *or*
 3. Urgency of defecation; *or*
 4. C-reactive protein (CRP); *or*
 5. Fecal calprotectin (FC); *or*
 6. Appearance of the mucosa on endoscopy, computed tomography enterography (CTE), magnetic resonance enterography (MRE), or intestinal ultrasound; *or*
 7. Improvement on a disease activity scoring tool (e.g., Ulcerative Colitis Endoscopic Index of Severity [UCEIS], Mayo score);

5. Immune checkpoint inhibitor-related toxicity

For all members (including new members) who meet all initial selection criteria.

4. Other

For all indications: Member has had a documented negative tuberculosis (TB) test (which can include a tuberculosis skin test [TST] or an interferon-release assay [IGRA]^{Footnote1*} within 12 months of initiating therapy for persons who are naïve to biologic drugs or targeted synthetic drugs associated with an increased risk of TB.

^{Footnote1*} If the screening testing for TB is positive, there must be further testing to confirm there is no active disease (e.g., chest x-ray). Do not administer the requested medication to members with active TB infection. If there is latent disease, TB treatment must be started before initiation of the requested medication.

For all indications: Member cannot use the requested medication concomitantly with any other biologic drug or targeted synthetic drug for the same indication.

5. Related Policies

1. CPB 0205 - Phototherapy and Photochemotherapy (PUVA) for Skin Conditions
2. CPB 0249 - Inflammatory Bowel Disease: Serologic Markers and Pharmacogenomic and Metabolic Assessment of Thiopurine Therapy
3. CPB 0315 - Etanercept
4. CPB 0341 - Infliximab
5. CPB 0577 - Laser Treatment for Psoriasis and Other Selected Skin Conditions
6. CPB 0655 - Adalimumab
7. CPB 0720 - Abatacept (Orencia)
8. CPB 0751 - Natalizumab
9. CPB 0761 - Certolizumab Pegol (Cimzia)
10. CPB 0790 - Golimumab (Simponi and Simponi Aria)
11. CPB 0885 - Vedolizumab (Entyvio)
12. CPB 0905 - Secukinumab (Cosentyx)
13. CPB 1009 - Risankizumab-rzaa (Skyrizi)
14. CPB 1011 - Guselkumab (Tremfya)

Dosage and Administration

Note: Approvals may be subject to dosing limits in accordance with FDA-approved labeling, accepted compendia, and/or evidence-based practice guidelines. Below includes dosing recommendations per the FDA-approved prescribing information. See Medical Specialty Medication Quantity Limits for more information.

Note: Ustekinumab is available in generic and biosimilar formulations. Some of the branded (proprietary name) biosimilar products below may be available in an unbranded equivalent. Furthermore, some ustekinumab biosimilars have received FDA-approval for interchangeable designation. Interchangeability designations allow for a biosimilar to be substituted for a reference product at the pharmacy level without having to obtain permission from a physician, depending on state pharmacy laws. This is similar to how generic drugs are routinely substituted for brand-name drugs. Not all biosimilars are interchangeable.

Stelara

Stelara is supplied in the following dosage forms and strengths:

- Subcutaneous (SC) Injection
 - Injection: 45 mg/0.5 mL or 90 mg/mL solution in a single-dose prefilled syringe
 - Injection: 45 mg/0.5 mL in a single-dose vial
 - In pediatrics, it is recommended that subcutaneous Stelara be administered by a healthcare provider. If a healthcare provider determines that it is appropriate, a person may self-inject or a caregiver may inject Stelara after proper training in subcutaneous injection technique;
- Intravenous (IV) Infusion

Injection: 130 mg/26 mL (5 mg/mL) solution in a single-dose vial.

Crohn's Disease and Ulcerative Colitis

- Intravenous (IV) induction adult dosage regimen:

A single IV infusion using weight-based dosing as per below:

- For adults weighing 55 kg or less, the recommended dosage is 260 mg (2 vials);
- For adults weighing more than 55 kg to 85 kg, the recommended dosage is 390 mg (3 vials);
- For adults weighing more than 85 kg, the recommended dosage is 520 mg (4 vials);

- Subcutaneous (SC) maintenance adult dosage regimen:

The recommended maintenance dosage is a subcutaneous 90 mg dose administered 8 weeks after the initial IV dose, then every 8 weeks thereafter.

Plaque Psoriasis

- Subcutaneous (SC) adult dosage regimen:
 - For persons weighing 100 kg or less, the recommended dose is 45 mg initially and 4 weeks later, followed by 45 mg every 12 weeks.
 - For persons weighing more than 100 kg, the recommended dose is 90 mg initially and 4 weeks later, followed by 90 mg every 12 weeks.
 - In subjects weighing more than 100 kg, 45 mg was also shown to be efficacious; however, 90 mg resulted in greater efficacy in these subjects;
- Subcutaneous (SC) pediatric dosage regimen:
 - Administered SC at Weeks 0 and 4, then every 12 weeks thereafter.
 - The recommended dose for pediatric persons (6-17 years old) is based on body weight:
 - body weight less than 60 kg, the recommended dose is 0.75 mg/kg
 - body weight 60 kg to 100 kg, the recommended dose is 45 mg
 - body weight more than 100 kg, the recommended dose is 90 mg.

Psoriatic Arthritis

- Subcutaneous (SC) adult dosage regimen:
 - The recommended dose is 45 mg initially and 4 weeks later, followed by 45 mg every 12 weeks.
 - For adults with co-existent moderate-to-severe plaque psoriasis weighing more than 100 kg, the recommended dose is 90 mg initially and 4 weeks later, followed by 90 mg every 12 weeks;
- Subcutaneous (SC) pediatric dosage regimen:
 - Administered (SC) at Weeks 0 and 4, then every 12 weeks thereafter.
 - The recommended dose for pediatric persons (6 to 17 years old) with psoriatic arthritis is based on body weight:
 - body weight less than 60kg, the recommended dose is 0.75 mg/kg
 - body weight 60mg or more, the recommended dose is 45 mg
 - body weight greater than 100 kg with co-existent moderate-to-severe plaque psoriasis, the recommended dose is 90 mg.

Source: Janssen Biotech, 2024

Imuldosa

Imuldosa is considered a biosimilar product to Stelara.

Imuldosa is supplied in the following dosage forms and strengths:

- Subcutaneous (SC) Injection
 - Injection: 45 mg/0.5 mL or 90 mg/mL solution in a single-dose prefilled syringe
 - In pediatrics, it is recommended that subcutaneous Imuldosa be administered by a healthcare provider. If a healthcare determines that it is appropriate, a person may self-inject or a caregiver may inject Imuldosa after proper training in subcutaneous injection technique;
- Intravenous (IV) Infusion

Injection: 130 mg/26 mL (5 mg/mL) solution in a single-dose vial.

Imuldosa carries the same labeled indications and dosing as Stelara except for the following indications:

Plaque Psoriasis

- Subcutaneous (SC) pediatric dosage regimen:
 - Administered SC at Weeks 0 and 4, then every 12 weeks thereafter.

- The recommended dose for pediatric persons (6-17 years old) is based on body weight:
 - body weight 60 kg to 100 kg, the recommended dose is 45 mg
 - body weight more than 100 kg, the recommended dose is 90 mg.
 - There is no dosage form for Imuldosa that allows weight-based dosing for pediatric persons below 60 kg.

Psoriatic Arthritis

- Subcutaneous (SC) pediatric dosage regimen:
 - Administered (SC) at Weeks 0 and 4, then every 12 weeks thereafter.
 - The recommended dose for pediatric persons (6 to 17 years old) with psoriatic arthritis is based on body weight:
 - body weight 60mg or more, the recommended dose is 45 mg
 - body weight greater than 100 kg with co-existent moderate-to-severe plaque psoriasis, the recommended dose is 90 mg
 - There is no dosage form for Imuldosa that allows weight-based dosing for pediatric persons below 60 kg.

Source: Accord BioPharma, 2024

Otulf

Otulf is considered a biosimilar product to Stelara.

Otulf is supplied in the following dosage forms and strengths:

- Subcutaneous (SC) Injection
 - Injection: 45 mg/0.5 mL or 90 mg/mL solution in a single-dose prefilled syringe
 - In pediatrics, it is recommended that subcutaneous Otulf be administered by a healthcare provider. If a healthcare determines that it is appropriate, a person may self-inject or a caregiver may inject Otulf after proper training in subcutaneous injection technique;
- Intravenous (IV) Infusion
 - Injection: 130 mg/26 mL (5 mg/mL) solution in a single-dose vial.

Otulf carries the same labeled indications and dosing as Stelara except for the following indications:

Plaque Psoriasis

- Subcutaneous (SC) pediatric dosage regimen:
 - Administered SC at Weeks 0 and 4, then every 12 weeks thereafter.
 - The recommended dose for pediatric persons (6-17 years old) is based on body weight:
 - body weight 60 kg to 100 kg, the recommended dose is 45 mg
 - body weight more than 100 kg, the recommended dose is 90 mg.
 - There is no dosage form for Otulf that allows weight-based dosing for pediatric persons below 60 kg.

Psoriatic Arthritis

- Subcutaneous (SC) pediatric dosage regimen:
 - Administered (SC) at Weeks 0 and 4, then every 12 weeks thereafter.
 - The recommended dose for pediatric persons (6 to 17 years old) with psoriatic arthritis is based on body weight:
 - body weight 60mg or more, the recommended dose is 45 mg
 - body weight greater than 100 kg with co-existent moderate-to-severe plaque psoriasis, the recommended dose is 90 mg
 - There is no dosage form for Otulf that allows weight-based dosing for pediatric persons below 60 kg.

Source: Fresenius Kabi USA, 2024

Pyzchiva

Pyzchiva is considered a biosimilar product to Stelara.

Pyzchiva is supplied in the following dosage forms and strengths:

- Subcutaneous (SC) Injection
 - Injection: 45 mg/0.5 mL or 90 mg/mL solution in a single-dose prefilled syringe
 - In pediatrics, it is recommended that subcutaneous Pyzchiva be administered by a healthcare provider. If a healthcare determines that it is appropriate, a person may self-inject or a caregiver may inject Pyzchiva after proper training in subcutaneous injection technique;
- Intravenous (IV) Infusion

Injection: 130 mg/26 mL (5 mg/mL) solution in a single-dose vial.

Pyzchiva carries the same labeled indications and dosing as Stelara except for the following indications:

Plaque Psoriasis

- Subcutaneous (SC) pediatric dosage regimen:
 - Administered SC at Weeks 0 and 4, then every 12 weeks thereafter.
 - The recommended dose for pediatric persons (6-17 years old) is based on body weight:
 - body weight 60 kg to 100 kg, the recommended dose is 45 mg
 - body weight more than 100 kg, the recommended dose is 90 mg.
 - There is no dosage form for Pyzchiva that allows weight-based dosing for pediatric persons below 60 kg.

Psoriatic Arthritis

- Subcutaneous (SC) pediatric dosage regimen:
 - Administered (SC) at Weeks 0 and 4, then every 12 weeks thereafter.
 - The recommended dose for pediatric persons (6 to 17 years old) with psoriatic arthritis is based on body weight:
 - body weight 60mg or more, the recommended dose is 45 mg
 - body weight greater than 100 kg with co-existent moderate-to-severe plaque psoriasis, the recommended dose is 90 mg
 - There is no dosage form for Pyzchiva that allows weight-based dosing for pediatric persons below 60 kg.

Source: Sandoz, 2024

Selarsdi

Selarsdi is considered a biosimilar product to Stelara.

Selarsdi is supplied in the following dosage forms and strengths:

- Subcutaneous (SC) Injection
 - Injection: 45 mg/0.5 mL or 90 mg/mL solution in a single-dose prefilled syringe
 - In pediatrics, it is recommended that subcutaneous Selarsdi be administered by a healthcare provider. If a healthcare determines that it is appropriate, a person may self-inject or a caregiver may inject Selarsdi after proper training in subcutaneous injection technique;
- Intravenous (IV) Infusion

Injection: 130 mg/26 mL (5 mg/mL) solution in a single-dose vial.

Selarsdi carries the same labeled indications and dosing as Stelara except for the following indications:

Plaque Psoriasis

- Subcutaneous (SC) pediatric dosage regimen:
 - Administered SC at Weeks 0 and 4, then every 12 weeks thereafter.
 - The recommended dose for pediatric persons (6-17 years old) is based on body weight:
 - body weight 60 kg to 100 kg, the recommended dose is 45 mg
 - body weight more than 100 kg, the recommended dose is 90 mg.
 - There is no dosage form for Selarsdi that allows weight-based dosing for pediatric persons below 60 kg.

Psoriatic Arthritis

- Subcutaneous (SC) pediatric dosage regimen:
 - Administered (SC) at Weeks 0 and 4, then every 12 weeks thereafter.
 - The recommended dose for pediatric persons (6 to 17 years old) with psoriatic arthritis is based on body weight:
 - body weight 60mg or more, the recommended dose is 45 mg
 - body weight greater than 100 kg with co-existent moderate-to-severe plaque psoriasis, the recommended dose is 90 mg
 - There is no dosage form for Selarsdi that allows weight-based dosing for pediatric persons below 60 kg.

Source: Alvotech USA, 2024

Starjemza

Starjemza is considered a biosimilar product to Stelara.

Starjemza is supplied in the following dosage forms and strengths:

- Subcutaneous (SC) Injection
 - Injection: 45 mg/0.5 mL or 90 mg/mL solution in a single-dose prefilled syringe; and 45 mg/0.5 mL in a single-dose glass vial
 - In pediatrics, it is recommended that subcutaneous Starjemza be administered by a healthcare provider. If a healthcare determines that it is appropriate, a person may self-inject or a caregiver may inject Starjemza after proper training in subcutaneous injection technique;
- Intravenous (IV) Infusion

Injection: 130 mg/26 mL solution supplied as a single-dose vial.

Starjemza carries the same labeled indications and dosing as Stelara.

Source: Hilka Pharmaceuticals USA, 2025

Steqeyma

Steqeyma is considered a biosimilar product to Stelara.

Steqeyma is supplied in the following dosage forms and strengths:

- Subcutaneous (SC) Injection
 - Injection: 45 mg/0.5 mL or 90 mg/mL solution in a single-dose prefilled syringe
 - In pediatrics, it is recommended that subcutaneous Steqeyma be administered by a healthcare provider. If a healthcare determines that it is appropriate, a person may self-inject or a caregiver may inject Steqeyma after proper training in subcutaneous injection technique;
- Intravenous (IV) Infusion

Injection: 130 mg/26 mL (5 mg/mL) solution in a single-dose vial.

Steqeyma carries the same labeled indications and dosing as Stelara except for the following indications:

Plaque Psoriasis

- Subcutaneous (SC) pediatric dosage regimen:
 - Administered SC at Weeks 0 and 4, then every 12 weeks thereafter.
 - The recommended dose for pediatric persons (6-17 years old) is based on body weight:
 - body weight 60 kg to 100 kg, the recommended dose is 45 mg
 - body weight more than 100 kg, the recommended dose is 90 mg.
 - There is no dosage form for Steqeyma that allows weight-based dosing for pediatric persons below 60 kg.

Psoriatic Arthritis

- Subcutaneous (SC) pediatric dosage regimen:
 - Administered (SC) at Weeks 0 and 4, then every 12 weeks thereafter.
 - The recommended dose for pediatric persons (6 to 17 years old) with psoriatic arthritis is based on body weight:
 - body weight 60mg or more, the recommended dose is 45 mg
 - body weight greater than 100 kg with co-existent moderate-to-severe plaque psoriasis, the recommended dose is 90 mg
 - There is no dosage form for Steqeyma that allows weight-based dosing for pediatric persons below 60 kg.

Source: Celltrion USA, 2024

Wezlana

Wezlana is considered a biosimilar product to Stelara.

Wezlana carries the same labeled indications and dosing as Stelara.

Source: Amgen, 2025

Yesintek

Yesintek is considered a biosimilar product to Stelara.

Yesintek carries the same labeled indications and dosing as Stelara.

Source: Biocon Biologics, 2024

Experimental, Investigational, or Unproven

Aetna considers concomitant use of ustekinumab with any other biologic drug (e.g., adalimumab, anakinra, etanercept, infliximab, rilonacept, tocilizumab) or targeted synthetic drug (e.g. tofacitinib) experimental, investigational, or unproven for the same indication because the effectiveness of this approach has not been established.

Aetna considers combined ustekinumab and vedolizumab for the treatment of inflammatory bowel disease (IBD) experimental, investigational, or unproven.

Aetna considers measurements of serum levels of ustekinumab and antibodies to ustekinumab (e.g., the Prometheus Anser UST test) experimental, investigational, or unproven because the clinical value of these measurements for members receiving ustekinumab therapy has not been established.

Aetna considers ustekinumab products experimental, investigational, or unproven for all other indications, including *any* of the following, because its effectiveness for these indications has not been established (not an all-inclusive list):

- Alopecia areata
- Alzheimer disease

- Atopic dermatitis
- Behcet's syndrome
- Biliary cholangitis (also known as biliary cirrhosis)
- Bullous pemphigoid
- Chronic antibiotic-refractory pouchitis
- Collagenous colitis (including refractory disease)
- Contact dermatitis
- Cutaneous lupus erythematosus,
- Eosinophilic dermatoses
- Erythrodermic psoriasis
- Graft versus host disease
- Guttate psoriasis
- Hidradenitis suppurativa or hidradenitis suppurativa sarcoidosis
- Large vessel vasculitis (e.g., giant cell arteritis and Takayasu arteritis)
- Lichen planus
- Leukocyte adhesion deficiency type 1
- Microscopic colitis (MC) (including steroid-refractory MC)
- Multiple sclerosis
- Pityriasis rubra pilaris
- Polymyalgia rheumatic
- Pyoderma gangrenosum
- Rheumatoid arthritis
- SAPHO syndrome
- Sarcoidosis
- Spondyloarthropathy
- Systemic lupus erythematosus,
- Uveitis
- Vitiligo.

Table:

CPT Codes / HCPCS Codes / ICD-10 Codes

CPT codes not covered for indications listed in the CPB:

Information in the [brackets] below has been added for clarification purposes. Codes requiring a 7th character are represented by "+":

Code	Code Description
Measurement of serum levels of ustekinumab and antibodies to ustekinumab -hyphen no specific code	
Other CPT codes related to the CPB :	
71045 – 71048	Radiologic examination, chest
86140	C-hyphenreactive protein
86141	C-hyphenreactive protein; high sensitivity (hsCRP)
86480	Tuberculosis test, cell mediated immunity antigen response measurement; gamma interferon
86481	enumeration of gamma interferon – producing T cells in cell suspension
86580	Skin test; tuberculosis, intradermal
96365 -hyphen 96368	Intravenous infusion, for therapy, prophylaxis, or diagnosis
96369 -hyphen 96371	Subcutaneous infusion for therapy or prophylaxis (specify substance or drug)
96372	Therapeutic, prophylactic, or diagnostic injection (specify substance or drug); subcutaneous or intramuscular
96379	Unlisted therapeutic, prophylactic, or diagnostic intravenous or intra-hyphenarterial injection or infusion
96401	Chemotherapy administration, subcutaneous or intramuscular; non-hyphenhormonal anti-hyphenneoplastic

Information in the [brackets] below has been added for clarification purposes. Codes requiring a 7th character are represented by "+":

Code	Code Description
96409 -hyphen 96411	intravenous, push technique, single or initial, or each additional
96413 -hyphen 96417	Chemotherapy administration; intravenous infusion technique

HCPSC codes covered if selection criteria are met:

J3357	Injection, ustekinumab, 1 mg
J3358	Ustekinumab, for intravenous injection, 1 mg
Q5098	Injection, ustekinumab-hyphensrlf (imuldosa), biosimilar, 1 mg
Q5099	Injection, ustekinumab-hyphenstba (steqeyma), biosimilar, 1 mg
Q5100	Injection, ustekinumab-hyphenkfce (yesintek), biosimilar, 1 mg
Q5137	Injection, ustekinumab-hyphenauub (wezlana), biosimilar, subcutaneous, 1 mg
Q5138	Injection, ustekinumab-hyphenauub (wezlana), biosimilar, intravenous, 1 mg
Q9996	Injection, ustekinumab-hyphenttwe (pyzchiva), subcutaneous, 1 mg
Q9997	Injection, ustekinumab-hyphenttwe (pyzchiva), intravenous, 1 mg
Q9998	Injection, ustekinumab-hyphenaekn (selarsdi), 1 mg
Q9999	Injection, ustekinumab-hyphenauuz (otulfi), biosimilar, 1 mg

Other HCPSC codes related to the CPB:

Mesalamine, balsalazide -hyphenno specific code

J0702	Injection, betamethasone acetate 3mg and betamethasone sodium phosphate 3mg
J0717	Injection, certolizumab pegol 1 mg [not covered with Ustekinumab]
J1020	Injection, methylprednisolone acetate, 20 mg
J1030	Injection, methylprednisolone acetate, 40 mg
J1040	Injection, methylprednisolone acetate, 80 mg
J1094	Injection, dexamethasone acetate, 1 mg
J1100	Injection, dexamethasone sodium phosphate, 1mg
J1438	Injection, etanercept, 25 mg
J1602	Injection, golimumab, 1mg for intravenous use [not covered with Ustekinumab]
J1700	Injection, hydrocortisone acetate, up to 25 mg
J1710	Injection, hydrocortisone sodium phosphate, up to 50 mg
J1720	Injection, hydrocortisone sodium succinate, up to 100 mg
J1745	Injection, infliximab, 10 mg [not covered with Ustekinumab]
J2650	Injection, prednisolone acetate, up to 1 ml
J2920	Injection, methylprednisolone sodium succinate, up to 40 mg
J2930	Injection, methylprednisolone sodium succinate, up to 125 mg
J3245	Injection, tildrakizumab, 1 mg
J3380	Injection, vedolizumab, 1 mg
J7500	Azathioprine, oral, 50 mg
J7501	Azathioprine, parenteral, 100 mg
J7506	Prednisone, oral, per 5 mg
J7509	Methylprednisolone, oral, per 4 mg
J7510	Prednisolone oral, per 5 mg
J7512	Prednisone, immediate release or delayed release, oral, 1 mg
J8540	Dexamethasone, oral, 0.25 mg
J8610	Methotrexate; oral, 2.5 mg
J8611	Methotrexate (jylamvo), oral, 2.5 mg

Information in the [brackets] below has been added for clarification purposes. Codes requiring a 7th character are represented by "+":

Code	Code Description
J8612	Methotrexate (xatmep), oral, 2.5 mg
J9250	Methotrexate sodium, 5 mg
J9255	Injection, methotrexate (accord) not therapeutically equivalent to j9250 or j9260, 50 mg
J9260	Methotrexate sodium, 50 mg
J7515	Cyclosporine, oral, 25 mg
J7516	Cyclosporine, parenteral, 250 mg
Q5109	Injection, infliximab-hyphenqbt, biosimilar, (ixifi), 10 mg
S0108	Mercaptopurine, oral, 50 mg

ICD-hyphen10 codes covered if selection criteria are met:

K50.00 -hyphen K50.919	Crohn's disease (regional enteritis)
K51.00 -hyphen K51.919	Ulcerative colitis
L40.0 -hyphen L40.3, L40.50 -hyphen L40.9	Psoriasis [not covered for erythrodermic psoriasis][age 6 and older]

ICD-hyphen10 codes not covered for indications listed in the CPB (not all inclusive):

Leukocyte adhesion deficiency type 1 -hyphen no specific code:

A15.0 -hyphen A15.9, A17.0 -hyphen A17.9, A18.01 -hyphen A18.89, A19.0 -hyphen A19.9	Tuberculosis [not covered for persons with active TB or untreated latent disease]
A36.89	Other diphtheritic complications [uveitis due to diphtheria]
A50.39	Other late congenital syphilitic ophthalmopathy [uveitis due to late congenital syphilis]
A51.43	Secondary syphilitic ophthalmopathy [uveitis due to syphilis]
A52.71	Late syphilitic ophthalmopathy [uveitis due to late syphilis]
A54.32	Gonococcal iridocyclitis
B02.32	Zoster iridocyclitis
B58.09	Other toxoplasma ophthalmopathy [uveitis due to toxoplasmosis]
D72.10 -hyphen D72.19	Eosinophilia [eosinophilic dermatoses]
D86.0 -hyphen D86.9	Sarcoidosis
D89.810 -hyphen D89.813	Graft-hyphenversus-hyphenhost disease
G30.0 -hyphen G30.9	Alzheimer's disease
G35	Multiple sclerosis
H20.00 -hyphen H20.9	Iridocyclitis
H44.131 -hyphen H44.139	Sympathetic uveitis
K52.831	Collagenous colitis
K52.838 – K52.839	Microscopic colitis
K74.3 -hyphen K74.5	Biliary cirrhosis
K91.850	Pouchitis [chronic antibiotic-hyphenrefractory]
L12.0	Bullous pemphigoid
L20.0 -hyphen L20.9	Atopic dermatitis
L23.0 – L25.9	Contact dermatitis
L40.4	Guttate psoriasis

Information in the [brackets] below has been added for clarification purposes. Codes requiring a 7th character are represented by "+":

Code	Code Description
L43.0 – L43.9	Lichen planus
L44.0	Pityriasis rubra pilaris
L63.0 -hyphen L63.9	Alopecia areata
L73.2	Hidradenitis suppurativa
L80	Vitiligo
L88	Pyoderma gangrenosum
L93.1	Subacute cutaneous lupus erythematosus
M05.00 -hyphen M06.9	Rheumatoid arthritis
M31.4	Aortic arch syndrome [Takayasu]
M31.5 -hyphen M31.6	Giant cell arteritis
M32.0 -hyphen M32.9	Systemic lupus erythematosus
M35.2	Behcet's disease
M35.3	Polymyalgia rheumatica
M45.0 -hyphen M49.89	Spondylopathies [spondyloarthropathy]
M86.30 -hyphen M86.39	Chronic multifocal osteomyelitis [SAPHO Syndrome]

Background

U.S. Food and Drug Administration (FDA)-Approved Indications

- Moderate to severe plaque psoriasis (PsO) in patients 6 years and older who are candidates for phototherapy or systemic therapy
- Active psoriatic arthritis (PsA) in patients 6 years and older
- Moderately to severely active Crohn's disease (CD) in adults
- Moderately to severely active ulcerative colitis (UC) in adults

Compendial Use

- Immune checkpoint inhibitor-related toxicity

Ustekinumab is a human interleukin-12 and -23 antagonist. IL-12 and IL-23 are naturally occurring cytokines that are involved in inflammatory and immune responses, such as natural killer cell activation and CD4+ T-cell differentiation and activation. In in vitro models, ustekinumab was shown to disrupt IL-12 and IL-23 mediated signaling and cytokine cascades by disrupting the interaction of these cytokines with a shared cell-surface receptor chain, IL-12Rβ1. The cytokines IL-12 and IL-23 have been implicated as important contributors to the chronic inflammation that is a hallmark of Crohn's disease and ulcerative colitis. In animal models of colitis, genetic absence or antibody blockade of the p40 subunit of IL-12 and IL-23, the target of ustekinumab, was shown to be protective (Janssen Biotech, 2024).

Ustekinumab carries labeled warnings and precautions for risk of infections, tuberculosis (TB), malignancies, hypersensitivity reactions, and noninfectious pneumonia. Serious bacterial, mycobacterial, fungal, and viral infections have been observed in patients receiving a ustekinumab product. Individuals genetically deficient in IL-12/IL-23 are particularly vulnerable to disseminated infections from mycobacteria (including nontuberculous, environmental mycobacteria), salmonella (including nontyphi strains), and Bacillus Calmette-Guerin (BCG) vaccinations. Serious infections and fatal outcomes have been reported in such patients. It is not known whether patients with pharmacologic blockade of IL-12/IL-23 from treatment with a ustekinumab product may be susceptible to these types of infections.

Patients are to be evaluated for tuberculosis (T infection prior to initiating ustekinumab therapy. Ustekinumab therapy should be avoided in patients with active TB infection. Treatment should be initiated for latent TB prior to administering a ustekinumab product. Anti-tuberculosis therapy should be considered prior to initiation of ustekinumab in patients with a past history of latent or active TB in whom an adequate course of treatment cannot be confirmed.

Per the Centers for Disease Control and Prevention (CDC), the tuberculin (TB) skin test (also called the Mantoux tuberculin skin test [TST]) and the TB blood test (also called interferon-gamma release assays or IGRAs) can be used to test for M. tuberculosis

infection. Additional tests are required to confirm TB disease. "A posterior-anterior chest radiograph is used to detect chest abnormalities. Lesions may appear anywhere in the lungs and may differ in size, shape, density, and cavitation. These abnormalities may suggest TB, but cannot be used to definitively diagnose TB. However, a chest radiograph may be used to rule out the possibility of pulmonary TB in a person who has had a positive reaction to a TST or TB blood test and no symptoms of disease" (CDC, 2024).

Since, ustekinumab is an immunosuppressant, it may increase the risk of malignancy. Malignancies were reported among patients who received ustekinumab in clinical trials. Hypersensitivity reactions, including anaphylaxis and angioedema, have also reported. In addition, two cases of posterior reversible encephalopathy syndrome (PRES), also known as Reversible Posterior Leukoencephalopathy Syndrome (RPLS), were reported in clinical trials. Cases have also been reported in post-marketing experience in patients with psoriasis, psoriatic arthritis, and Crohn's disease.

Prior to initiating therapy with ustekinumab, it is recommended that patients receive all age-appropriate immunizations per current immunization guidelines. Patients being treated with ustekinumab should avoid receiving live vaccines.

The most common adverse reactions include the following:

- Psoriasis (3% or more): nasopharyngitis, upper respiratory tract infection, headache, and fatigue.
- Crohn's Disease, induction (3% or more): vomiting.
- Crohn's Disease, maintenance (3% or more): nasopharyngitis, injection site erythema, vulvovaginal candidiasis/mycotic infection, bronchitis, pruritus, urinary tract infection, and sinusitis.
- Ulcerative colitis, induction (3% or more): nasopharyngitis .
- Ulcerative colitis, maintenance (3% or more): nasopharyngitis, headache, abdominal pain, influenza, fever, diarrhea, sinusitis, fatigue, and nausea.

In 2009, Stelara (Janssen Biotech, Inc) became the first brand of ustekinumab that was FDA-approved.

In October 2023, Wezlana (Amgen, Inc.) was FDA-approved as the first interchangeable biosimilar to Stelara for the same indications and dosage. Approval was based on a comprehensive review of scientific evidence demonstrating it is highly similar to Stelara and that there are no clinically meaningful differences between the two products in terms of safety and efficacy. This evidence included comparisons of the products on an analytical level using an extensive battery of chemical and biological tests and biological assays that confirmed similarity in the structural and functional features of Wezlana and Stelara, comparative human pharmacokinetic data, clinical immunogenicity data, and other clinical safety and effectiveness data. The evidence also demonstrated that Wezlana met the other legal requirements to be interchangeable with Stelara at the pharmacy level (FDA, 2023).

In April 2024, the FDA approved Stelara biosimilar Selarsdi (ustekinumab-aekn) (Alvotect USA Inc).

In June 2024, the FDA approved Stelara biosimilar Pyzchiva (ustekinumab-ttwe) (Sandoz Inc.).

In September 2024, the FDA approved Stelara biosimilar Otulfi (ustekinumab-aaaz) (Fresenius Kabi USA, LLC).

In October 2024, the FDA approved Stelara biosimilars Imuldosa (ustekinumab-srlf) (Accord BioPharma Inc.) and unbranded ustekinumab-aekn (Alvotect USA Inc.).

In December 2024, the FDA approved Stelara biosimilars Steqeyma (ustekinumab-stba) (Celltrion USA, Inc) and Yesintek (ustekinumab-kfce) (Biocon Biologics Inc.).

In March 2025, the FDA approved Stelara biosimilar ustekinumab-ttwe (Quallent Pharmaceuticals Health LLC), unbranded formulation of Pyzchiva (Sandoz Inc.).

In April 2025, the FDA approved unbranded ustekinumab (Janssen Biotech, Inc.) for the same indications as its predicate, Stelara (Janssen Biotech, Inc.).

In May 2025, the FDA granted approval for ustekinumab-aaaz (Otulfi) as an interchangeable biosimilar to the reference product Stelara. Otulfi is now available as unbranded ustekinumab-aaaz (Fresenius Kabi USA, LLC). In addition, Steqeyma is now available as unbranded ustekinumab-stba (Celltrion USA, Inc.). Moreover, the FDA granted approval of ustekinumab-hmny (Starjemza) (Hikma Pharmaceuticals USA Inc.) as another biosimilar referencing Stelara.

Biosimilar and Interchangeable Drugs

Biosimilar drugs are FDA-approved biological products that have demonstrated that they are highly similar without clinically meaningful differences in terms of safety and effectiveness from the biologic reference product. Only minor differences in clinically inactive components are allowable in biosimilar drugs. Some biosimilar drugs may have the 'interchangeable' designation. An interchangeable biological product is a biosimilar that meets additional FDA requirements and may be substituted for the reference product at the pharmacy without the intervention of the prescribing health care provider, similar to

how generic drugs are routinely substituted for brand-name drugs (FDA, 2023; FDA, undated). For approval as an interchangeable product, the biosimilar must be expected to produce the same clinical result as the reference product in any given patient, and for a product administered more than once, the risk of switching between a reference product and an interchangeable product must not be greater than the risk of using the reference product without switching (Herndon et al, 2023).

Despite the adoption of biosimilars in many therapeutic areas, concerns persist regarding switching a patient to a biosimilar whose condition is stable while on the reference biologic. Herndon et al (2023) performed a systematic review and meta-analysis to examine the occurrence of safety events following a switch to or from a biosimilar and its reference biologic in all identified controlled clinical studies that included a biosimilar approved by the FDA. The authors reviewed FDA databases containing publicly available information for all randomized controlled studies and extension studies with a switch treatment period (STP) to or from a biosimilar and its reference biologic. Information not captured in FDA reviews was identified by searching the Embase, MEDLINE, and PubMed databases for the period of January 1, 2000, through December 31, 2022. Forty-four STPs were identified from 31 unique studies for 21 different biosimilars. Data were extracted and synthesized following PRISMA guidelines. Meta-analysis was conducted to estimate the overall risk difference across studies. A total of 5,252 patients who were switched to or from a biosimilar and its reference biologic were identified. Safety data including deaths, serious adverse events (SAEs), and treatment discontinuation showed an overall risk difference (95% CI) of -0.00 (-0.00, 0.00), 0.00 (-0.01, 0.01), -0.00 (-0.01, 0.00) across STPs, respectively. Immunogenicity data showed similar incidence of anti-drug antibodies and neutralizing antibodies in patients within a STP who were switched to or from a biosimilar to its reference biologic and patients who were not switched. Immune related adverse events such as anaphylaxis, hypersensitivity reactions, and injections site reactions were similar in switched and non-switched patients. The authors report that this is the first systematic review using statistical methods to address the risk of switching patients between reference biologics and biosimilars which found no difference in the safety profiles or immunogenicity rates in patients who were switched and those who remained on a reference biologic or a biosimilar. It is of note that endpoints related to pharmacokinetics (PK) or effectiveness of a drug that were collected during the STPs were not included in this systematic review. If efficacy data were collected periodically during STPs, it was not powered for an efficacy determination. The efficacy data used to determine equivalent efficacy of a biosimilar to a reference biologic is contained in publicly available FDA reviews and in the publications for the individual studies that led to approval of the biosimilars.

In summary, all biological products are approved only after they meet the FDA's rigorous approval standards. This means health care providers and patients can expect the same safety and effectiveness from both a biosimilar and an interchangeable biosimilar, just as they would for a reference product (FDA, 2023; FDA, undated).

The FDA previously recommended that Prescribing Information (labeling) include a "Biosimilarity Statement" or "Interchangeability Statement" describing the product's relationship to its reference product in the Highlights section. However, in 2024, the FDA recommended that all labeling for biosimilars, including interchangeable biosimilars, include one statement - a "biosimilarity statement". The FDA state that statements in the Prescribing Information identifying that products have been approved as interchangeable with the reference product and describing the interchangeability standard are not necessary for informing the safe and effective use of the product to prescribing health care professionals. Moreover, there may be inaccurate perceptions that interchangeable biosimilars are safer or more effective than biosimilars that are not approved as interchangeable. However, the interchangeability designation does not indicate a higher level of biosimilarity. Health care professionals can prescribe both biosimilar and interchangeable biosimilar products in place of the reference product with equal confidence that they are as safe and effective as their reference products. Thus, including biosimilarity statements in the Prescribing Information for both biosimilar and interchangeable biosimilar products may avoid confusion among health care professionals about the significance of FDA's determination that a biosimilar is interchangeable with the reference product (Yim, 2024).

"FDA recommends that biosimilar and interchangeable biosimilar product labeling incorporate relevant data and information from the reference product labeling, with appropriate modifications. The relevant data and information from the reference product labeling that should be incorporated into the biosimilar and interchangeable biosimilar product labeling will depend on whether the applicant is seeking licensure for all conditions of use (e.g., indication(s), dosing regimen(s)) or fewer than all conditions of use for which the reference product has been previously licensed. In addition, biosimilar and interchangeable biosimilar product labeling may include information specific to the biosimilar or interchangeable biosimilar product that is necessary to inform safe and effective use, including preparation, administration, and storage conditions. This information may differ from that of the reference product labeling to reflect differences between the biosimilar or interchangeable biosimilar product and the reference product" (Yim, 2024).

Biosimilar product labeling may include fewer indications than the reference product if the reference product has exclusivity rights for an indication. This can be due to patent rights or other unexpired exclusivity rights. However, "Prescribers can prescribe both biosimilar and interchangeable biosimilar products in place of the reference product with equal confidence that they are as safe and effective as their reference products" (Yim, 2024).

Alzheimer Disease

Husein-EIAhmed and Steinhoff (2022) noted that AD is a common chronic inflammatory condition. Ustekinumab targets the IL-12 and IL-23, and it may also play a role in AD. Administration of ustekinumab in AD is presented in anecdotal reports with conflicting results. In a systematic review, these investigators examined the clinical value of ustekinumab in the treatment of

patients with AD. They examined published data entailing AD patients treated with ustekinumab. The main outcome was clinical improvement. These researchers classified this in 3 categories: "complete response", "partial response", and "no response.". A multi-variant model was used to evaluate the effectiveness and the following potential predictive factors: gender, age, duration of AD, asthma, previous use of biologics, previous systemic therapies, levels of IgE and duration of ustekinumab therapy. Data on 23 patients were analyzed. Complete AD remission was reported in 8 patients (34.8 %), while the absence of response was observed also in 8 patients (34.8 %). Partial response was reported in 7 patients (30.4 %). No differences were observed with the predictive factors. The authors concluded that the IL-12/23 pathway is unlikely to be an attractive target for AD. Moreover, these investigators stated that more novel and effective treatments for AD are available and should be prioritized. The impact of anti-IL-12/IL-23 therapy in AD is still unclarified due to limited controlled trials.

Alopecia Areata

Renert-Yuval and Guttman-Yassky (2017) stated that alopecia areata (AA) lacks FDA-approved therapeutics for extensive cases, which are associated with very poor rates of spontaneous hair regrowth and major psychological distress. Current treatments for severe cases include broad immune-suppressants, which are associated with significant adverse effects, precluding long-term use, with rapid hair loss following treatment termination. As a result of the extent of the disease in severe cases, topical contact sensitizers and intralesional treatments are of limited use. The pathogenesis of AA is not yet fully understood, but recent investigations of the immune activation in AA skin reveal Th1/IFN- γ , as well as Th2, PDE4, IL-23, and IL-9 up-regulations. Tissue analyses of both animal models and human lesions following broad-acting and cytokine-specific therapeutics (such as JAK inhibitors and ustekinumab, respectively) provided another opportunity for important insights into the pathogenesis of AA. These investigators reviewed novel therapeutics that are undergoing clinical trials for AA, emphasizing the potential transformation of the clinical practice of AA, which is currently lacking. Moreover, they noted that clinical trials of broad-acting T cell antagonists (JAK inhibitors, PDE4 inhibitor), Th2 antagonists (tralokinumab), and IL-23/IL-17 antagonist (ustekinumab) are ongoing.

Atopic Dermatitis

Khatti and associates (2017) noted that atopic dermatitis (AD) is the most common inflammatory skin disease, but treatment options for moderate-to-severe disease are limited. In a double-blind, placebo-controlled phase-II clinical trial, these researchers evaluated the safety and effectiveness of ustekinumab in patients with moderate-to-severe AD. A total of 33 patients with moderate-to-severe AD were randomly assigned to either ustekinumab (n = 16) or placebo (n = 17), with subsequent cross-over at 16 weeks, and last dose at 32 weeks. Background therapy with mild topical steroids was allowed to promote retention. Study end-points included clinical (SCORAD50) and biopsy-based measures of tissue structure and inflammation, using protein and gene expression studies. The ustekinumab group achieved higher SCORAD50 responses at 12, 16 (the primary end-point) and 20 weeks compared to placebo, but the difference between groups was not significant. The AD molecular profile/transcriptome showed early robust gene modulation, with sustained further improvements until 32 weeks in the initial ustekinumab group. Distinct and more robust modulation of Th1, Th17 and Th22 but also Th2-related AD genes was seen after 4 weeks of ustekinumab treatment (i.e., MMP12, IL-22, IL-13, IFN- γ , elafin/PI3, CXCL1 and CCL17; $P < .05$). Epidermal responses (K16, terminal differentiation) showed faster (4 weeks) and long-term regulation (32 weeks) from baseline in the ustekinumab group. No severe adverse events (AEs) were observed. The authors concluded that ustekinumab had clear clinical and molecular effects, but clinical outcomes might have been obscured by a profound "placebo" effect, most likely due to background topical glucocorticoids and possibly insufficient dosing for AD.

Weiss and co-workers (2017) evaluated the immunologic effects of ustekinumab treatment on AD skin and correlated them with the effectiveness of this drug. These researchers examined the course of 3 patients with severe AD who were administered 45-mg of subcutaneous ustekinumab over a period of 16 weeks. Clinical scores and skin biopsy specimens, taken at baseline and at week 8, were used to assess changes in disease severity. All 3 patients showed a gradual improvement of the disease, achieving a 50 % reduction in the Eczema Area and Severity Index (EASI) score by week 16. Immunohistology of skin biopsy specimens revealed a significant decrease in the degree of epidermal hyperplasia/proliferation and the number of infiltrating dermal T cells, dendritic cells, and mast cells after treatment. Using quantitative real-time polymerase chain reaction (PCR) of lesional skin, these researchers found a clear reduction of T-helper 2-/22-associated molecules after therapy. The authors concluded that blocking IL-12/-23 p40 could be beneficial for a subgroup of patients with severely infiltrated AD. Moreover, they stated that the small number of patients (n = 3) limited efficacy analysis and warrants prospective placebo-controlled studies in larger patient cohorts.

In a randomized, double-blind, placebo-controlled, phase-II clinical trial, Saeki and colleagues (2017) evaluated the safety and effectiveness of ustekinumab in patients with severe AD. Patients (aged 20 to 65 years) with severe or very severe AD entered a 12-week, double-blind treatment period during which they received (1 : 1 : 1) ustekinumab 45-mg, 90-mg or placebo subcutaneous injections at weeks 0 and 4, with follow-up until week 24. The primary efficacy end-point was percentage change from baseline in EASI score at week 12. Major secondary efficacy end-points included the proportion of patients achieving EASI 50, EASI 75, Investigator's Global Assessment score 0 to 1, change from baseline Atopic Dermatitis Itch Scale and Dermatology Life Quality Index. A total of 79 patients were randomized [ustekinumab 45-mg (n = 24), 90-mg (n = 28), placebo (n = 27)]. Ustekinumab treatment showed non-significant improvement in least square mean change from baseline EASI score at week 12 [45-mg: -38.2 %, 95 % CI: -21.02 to 19.51; $p < 0.94$ and 90-mg: -39.8 %, 95 % CI: -21.84 to 17.14; $p < 0.81$] versus placebo (-37.5 %). A non-significant improvement in major secondary efficacy end-points was observed in both ustekinumab groups

versus placebo. The most common treatment-emergent AEs were nasopharyngitis and worsened AD (higher in placebo versus ustekinumab groups). The authors concluded that ustekinumab 45-mg and 90-mg did not demonstrate meaningful effectiveness in Japanese patients with severe AD; the treatment was generally well-tolerated.

Behcet's Syndrome

Vitale and associates (2017) noted that Behcet's syndrome (BS) is a systemic inflammatory disorder characterized by a wide range of potential clinical manifestations with no gold-standard therapy. However, the recent classification of BS at a crossroads between autoimmune and auto-inflammatory syndromes has paved the way to new further therapeutic opportunities in addition to anti-TNF agents. These investigators provided a digest of all current experience and evidence about pharmacological agents recently described as having a role in the treatment of BS, including IL-1 inhibitors, tocilizumab, rituximab, alemtuzumab, ustekinumab, interferon (IFN)-alpha-2a, and apremilast. IL-1 inhibitors currently represent the most studied agents among the latest treatment options for BS, proving to be effective, safe and with an acceptable retention on treatment. However, since BS is a peculiar disorder with clinical features responding to certain treatments that in turn can worsen other manifestations, identifying new therapeutic options for patients unresponsive to the current drug armamentarium is of great relevance. The authors noted that a number of agents have been studied in the past 10 years showing changing fortunes in some cases and promising results in others.

Biliary Cholangitis (Biliary Cirrhosis)

Floreani and Mangini (2018) noted that primary biliary cholangitis (PBC), formerly called primary biliary cirrhosis, is a chronic cholestatic liver disease that progresses slowly to end-stage liver disease. The 1st Food and Drug Administration (FDA)-approved treatment for PBC was ursodeoxycholic acid (UDCA). This treatment slows the progress of the disease, but about 30 to 40 % of patients fail to respond to UDCA. A number of options are under investigation as 2nd-line treatment. Obeticholic acid (OCA), a Farnesoid X receptor agonist, has been approved in May 2017 by the FDA for patients who are non-responders or intolerant to UDCA. The results of a randomized, double-blind, phase-III clinical trial of OCA compared to placebo, showed that approximately 50 % of patients reached a significant reduction in serum alkaline phosphatase, a marker predictive of disease progression, liver transplantation or death. Other emerging therapies include: agents targeting fibrosis, inflammation, or immunological response. Indeed, after 30 years of UDCA therapy as unique choice for PBC patients, a number of targets, derived from a deeper knowledge of the pathophysiology of the disease, has been discovered and they offer different and new therapeutic approaches that are now under evaluation; and ustekinumab was one of the keywords listed in the abstract of this study.

Chronic Antibiotic-Refractory Pouchitis

Ollech and colleagues (2019) noted that chronic antibiotic-refractory pouchitis (CARP) occurs in up to 15 % of patients with ulcerative colitis (UC) following proctocolectomy with ileal pouch-anal anastomosis (IPAA). In a retrospective single-center study, these investigators examined the effectiveness of ustekinumab in the treatment of CARP. This trial included UC patients with an IPAA, who subsequently developed CARP and received ustekinumab with standard CD dosing between 2016 and 2018. Patients with CD of the pouch were excluded. Demographic, clinical, and endoscopic data were collected. Outcomes included a change in the endoscopic subscore of the Pouchitis Disease Activity Index (PDAI), change in the ulcerated surface area, clinical response, and the number of bowel movements per 24 hours. A total of 24 patients with CARP were included for analysis. Median follow-up time was 12.9 months (inter-quartile range [IQR] 7.9 to 16); 12 patients (50 %) had a clinical response with the median number of bowel movements within 24 hours decreasing from 8 (IQR, 5 to 12) to 6 (IQR, 5 to 8); $p = 0.002$; 13 patients had pouchoscopies available post-ustekinumab treatment. In these patients, the median endoscopic subscore of the PDAI decreased from 5 (IQR, 3 to 6) to 4 (IQR, 2 to 5); $p = 0.016$. Likewise, among these 13 patients, 9 (69 %) had an ulcerated surface area greater than 10 % before ustekinumab treatment; after treatment with ustekinumab, only 4 patients (31 %) still had an ulcerated surface area of greater than 10 %. The authors concluded that this was the largest study of ustekinumab treatment for patients with CARP; ustekinumab therapy led to the improvement in clinical and endoscopic endpoints. This was a small ($n = 24$), retrospective study; its findings need to be validated by well-designed studies.

Furthermore, an UpToDate review on "Pouchitis: Management" (Shen, 2019) states that "Other agents have been used for treating chronic antibiotic refractory pouchitis, although data are limited. Vedolizumab, an anti-integrin antibody, and ustekinumab, an anti-interleukin 12/23 antibody appear to be effective for some patients with chronic antibiotic refractory pouchitis (CARP)".

Rocchi et al (2022) noted that CD of the pouch and chronic pouchitis represent the most common long-term complications of total proctocolectomy and ileal pouch anal anastomosis (IPAA) for the treatment of refractory UC. These conditions are treated with multiple agents, including antibiotics, immunomodulators, and biologics. Among the latter, ustekinumab is approved for both CD and UC. These investigators carried out a systematic review to examine the effectiveness of this anti-IL12/23 in CD of the pouch and chronic refractory pouchitis. PubMed, Embase, Ovid, and the Cochrane Controlled Trials Register were searched to identify studies published until August 2020 examining the use of ustekinumab for these conditions. A total of 86 eligible patients with IPAA -- 51 with CD of the pouch, 35 with chronic pouchitis -- were identified from 2 retrospective studies and 5 case reports. Reported clinical response to ustekinumab was 63 % and 85 % in chronic pouchitis and CD of the pouch after 4 to 12 weeks and

4 to 16 weeks, respectively. Clinical remission was reported in 10 % of patients with chronic pouchitis and 27 % of patients with CD of the pouch after 8 to 52 weeks and 4 to 52 weeks of treatment, respectively. Endoscopic response was reported in 60 % and 67 % of patients with chronic pouchitis and CD of the pouch after 24 to 32 weeks and 8 to 24 weeks of treatment, respectively. The authors concluded that there is a limited and inconclusive body of evidence suggesting that ustekinumab may be a therapeutic option for patients with chronic pouchitis and CD of the pouch refractory to other therapies. Moreover, these researchers stated that small sample sizes and large heterogeneity of therapy protocols/outcome definitions were the major drawbacks of this systematic review.

Crohn's Disease

Sandborn et al (2008) performed a double-blind, cross-over trial of the clinical effects of UST in 104 patients with moderate-to-severe CD. Patients were given subcutaneous placebo at weeks 0-3, then UST at weeks 8-11; subcutaneous UST at weeks 0-3, then placebo at weeks 8-11; intravenous placebo at week 0, then UST at week 8; or intravenous UST at week 0, then placebo at week 8. Clinical rates for the combined groups given UST and placebo were 53% and 30% ($p = .02$), respectively at weeks 4 and 6, and 49% and 40% ($p = .34$), respectively at week 8. In a subgroup of 49 patients who were previously given infliximab (neither primary nor secondary nonresponders), clinical response to UST was significantly greater than the group given placebo ($p < .05$) through week 8. There was no increase in the number of adverse or serious adverse events in patients given UST through week 8 compared with placebo. Sandborn et al concluded that UST induced a clinical response in patients with moderate-to-severe CD, especially in patients previously given infliximab.

Toedter et al (2009) studied UST induction therapy in a placebo-controlled trial of patients with CD ($n=104$). Among study subjects receiving UST, 49% achieved clinical response at week 8 versus 40% for placebo ($p=.34$). Further, in a subgroup of patients previously treated with infliximab ($n=49$), 59% receiving UST had a clinical response versus 26% receiving placebo ($p=.02$). The investigators found that mean changes from baseline C-reactive protein (CRP) at week 8 in the primary group were -0.3 and -3.1 mg/l after treatments with placebo ($n=43$) and UST ($n=46$), respectively ($p=.074$) while in the infliximab-experienced subgroup, the mean changes were +2.0 (placebo, $n=23$) and -2.6 mg/l (UST, $n=20$) ($p=.004$). Toedter et al concluded that the potential benefit of UST in CD is supported by serum CRP reductions and that evidence suggests that increased systemic inflammation as manifested by higher baseline CRP values leads to larger treatment effects with UST, particularly in infliximab-experienced patients.

Sandborn et al (2012) evaluated UST in adults with moderate-to-severe CD that was resistant to anti-tumor necrosis factor (TNF) treatment. The investigators randomly assigned 526 patients to receive intravenous ustekinumab (at a dose of 1, 3, or 6 mg per kilogram of body weight) or placebo at week 0, with the primary endpoint being clinical response at 6 weeks. During the maintenance phase, 145 patients who had a response to UST at 6 weeks underwent a second randomization to receive subcutaneous injections of UST (90 mg) or placebo at weeks 8 and 16. The proportions of patients who reached the primary endpoint were 36.6%, 34.1%, and 39.7% for 1, 3, and 6 mg of UST per kilogram, respectively, as compared with 23.5% for placebo ($p=.005$ for the comparison with the 6-mg group); the rate of clinical remission with the 6-mg dose did not differ significantly from the rate with placebo at 6 weeks. The investigators found that maintenance therapy with UST resulted in significantly increased rates of clinical remission (41.7% vs. 27.4%, $p=.03$) and response (69.4% vs. 42.5%, $p<.001$) at 22 weeks compared to placebo. The investigators concluded that for patients with moderate-to-severe CD resistant to TNF antagonists there was an increased rate of response to induction with UST compared with placebo and that patients with an initial response to ustekinumab had significantly increased rates of response and remission with UST as maintenance therapy.

A 2015 Cochrane review reported that UST and briakinumab (ABT-874) are monoclonal antibodies that target the standard p40 subunit of the cytokines interleukin-12 and interleukin-23 (IL-12/23p40). In this review methodological quality was assessed using the Cochrane risk of bias tool. The primary outcome was failure to induce clinical remission, defined as a CD activity index (CDAI) of < 150 points, while secondary outcomes included failure to induce clinical improvement, serious adverse events, and withdrawals due to adverse events. Clinical improvement was defined as decreases of > 70 or > 100 points in the CDAI from baseline with analysis on an intention-to-treat basis. Four randomized controlled trials ($n = 955$ patients) met their inclusion criteria. A low risk of bias was assigned to all studies. Two of these studies were ustekinumab studies (630 patients) and they were pooled despite differences in intravenous doses (i.e. 1mg/kg, 3 mg/kg, 4.5 mg/kg, and 6 mg/kg), however, the subcutaneous dose group was not included in the analysis, as it was unclear if subcutaneous was equivalent to intravenous dosing. At Week 6, 85% (356/420) of UST patients failed to enter remission compared to 89% (142/159) of placebo patients (RR 0.94, 95% confidence interval [CI] 0.88 to 1.01). Subgroup analysis showed no statistically significant difference by dose, but there were statistically significant differences in clinical improvement between UST and placebo-treated patients. In the UST group, 55% (230/420) of patients failed to improve clinically with a 70-point decline in CDAI score compared to 72% (115/159) of placebo patients (RR 0.75, 95% CI 0.66 to 0.86). Subgroup analysis revealed significant differences compared to placebo for the 1 mg/kg, 4.5 mg/kg and 6 mg/kg dosage subgroups. Similarly for a 100-point decline in CDAI, 62% (262/420) of patients in the UST group failed to improve clinically compared to 78% (124/159) of placebo patients (RR 0.79, 95% CI 0.71 to 0.89). Subgroup analysis showed a significant difference compared to placebo for the 4.5 mg/kg dose group. GRADE analyses of the UST studies rated the overall quality of the evidence for the outcomes for clinical remission and clinical response as moderate. There were no statistically significant differences in the incidence of adverse events, serious adverse events or withdrawal due to adverse events. Sixty-seven percent (316/473) of UST patients developed at least one adverse event compared to 73% (135/184) of placebo patients (RR 0.92, 95% CI 0.83 to 1.03). The authors concluded that although they were

uncertain about the efficacy of UST for induction of remission, moderate quality evidence suggested that UST may be effective for induction of clinical improvement in patients with moderate to severe CD (Khanna et al, 2015).

Khorrani et al (2016) assessed the effectiveness and safety of UST in refractory patients with CD in actual practice. The investigators recruited 116 consecutive patients with CD who were treated with subcutaneous UST between March 2010 and December 2014 were retrospectively included in a multicenter open-label study, with clinical response assessed after the loading doses, 6 months, 12 months, and last follow-up. Median follow-up was 10 months with an interquartile range of 5-21 months. Clinical response after loading UST was achieved in 97/116 (84%) patients while the clinical benefit at 6 months, 12 months, and at the end of the follow-up period was 76%, 64%, and 58%, respectively. Dose escalation was effective in 8 of 11 (73%) patients and perianal disease improved in 11 of 18 (61%) patients with active perianal fistulae. Adverse events, none of which required UST withdrawal, were reported in 11 (9.5%) patients. Khorrani et al concluded that subcutaneous ustekinumab is effective and safe in a high proportion of patients with CD that were resistant to conventional immunosuppressant and antitumor necrosis factor drugs.

Sandborn et al (2016) stated that interleukins 12 & 23 are linked to the pathophysiology of CD, wherein the pro-inflammatory cytokines are blocked by UST. They conducted a phase 3 trial to study IV UST in these patients. The study included 741 patients with moderate to severe CD (CDAI 220-450) who had failed or were intolerant to at least 1 TNF antagonist. Study subjects were randomized 1:1 to a single dose of placebo, 130 mg of UST, or a weight-based, tiered UST dose. The primary endpoint was clinical response at Week 6, with clinical response defined as reduction of CDAI score of more than 100 points. The study protocol also defined patients with a baseline CDAI between 220 and 248 points as having a clinical response with a CDAI score less than 150. At Week 8 patients were either transferred to a maintenance study or followed until Week 20.

Results of the study by Sandborn et al (2016) showed that Week 6 clinical response was observed in 33.7 % of subjects who received the 6 mg/kg UST dose and 34.3% of the 130 mg UST group versus 7.3% of the IV placebo group ($p = .003$ and $.002$, respectively). The investigators found that clinical remission occurred in 20.9% of the 6 mg/kg UST dose and 15.9% of the 130 mg UST dose, as compared with 7.3% of the placebo group ($p < 0.001$, $p = 0.003$). Clinical response at week 8 was seen in 37.8% of the 6 mg / kg UST dose group and 33.5% of the 130 mg UST group compared with 30.2% in the placebo group ($p \leq 0.001$ for each experimental group compared with placebo). The IV UST groups both had results of induction doses significantly improve CDAI, IBDQ, CRP, fecal lactoferrin, and calprotectin as compared with the placebo group. It is of note that no malignancies, deaths, major adverse cardiovascular events, or TB occurred in UST-treated patients through week 20. Sandborn et al concluded that among moderate to severe CD patients who are refractory to one or more TNF-antagonists, IV UST induced clinical response and remission and was well tolerated throughout induction.

Particularly in light of the results of the 2016 phase 3 study by Sandborn et al, the Food and Drug Administration has approved the use of ustekinumab for moderate to severe Crohn's disease.

For adult outpatients with moderate to severe Crohn's disease, guidelines from the American Gastroenterological Association (AGA) recommend induction and maintenance therapy with anti-tumor necrosis factor- α agents or ustekinumab over no treatment (Feuerstein et al, 2021).

Cutaneous Lupus Erythematosus

Romero-Mate and colleagues (2017) reported on the case of a 52-year old woman with a 28-year history of disfiguring facial discoid lupus erythematosus (DLE), persistent despite both classical therapies and rituximab. Ustekinumab 45-mg was started in combination with methotrexate and intralesional corticosteroids. Methotrexate and intralesional corticosteroids were withdrawn 30 months later and ustekinumab maintained as monotherapy; 48 months later stable improvement was achieved without side effects. The authors noted that only 9 patients with cutaneous lupus erythematosus (CLE) treated with ustekinumab had been reported to-date. They concluded that ustekinumab could be a promising alternative in severe and recalcitrant cases of CLE. These preliminary findings need to be validated in future well-designed studies.

Eosinophilic Dermatoses

Rozenblat and Ziv (2020) stated that eosinophilic dermatoses syndromes are rare diseases with a prominent eosinophilic infiltration mechanism. These syndromes have several known treatments with limited success. Several physicians world-wide suggested possible advantages of using specific biological drugs, which are different from eosinophil targeted biotherapies as treatments for eosinophilic dermatoses syndromes. Others considered these drugs as possible triggers. In a systematic review, these investigators examined the pros and cons of biological drugs as treatments and triggers of eosinophilic dermatoses.

Studies published in the last 30 years containing relevant key words were reviewed using PubMed and Medline. Associations between infliximab, adalimumab, etanercept, TNF- α inhibitors and ustekinumab to eosinophilic dermatoses syndromes were reviewed. The search revealed an association between 17 eosinophilic dermatoses patients and the drugs of interest. Out of 5 Wells' syndrome cases, 4 patients had an outbreak of the disease following treatment and 1 improved by the treatment; 6 cases of eosinophilic fasciitis mostly had a positive reaction to the treatment. More associations were found among 4 cases of Churg-Strauss syndrome, 1 case of granuloma faciale and 1 case of eosinophilic pustular folliculitis. The authors concluded that TNF- α inhibitors and ustekinumab may have a role in the treatment of eosinophilic dermatosis syndromes. These drugs may act as triggers among Wells' syndrome patients. Moreover, these researchers stated that further investigation is needed.

Guttate Psoriasis

In a case-series study, Brummer and co-workers (2017) reported their findings on the use of ustekinumab for the treatment of patients with refractory guttate psoriasis (n = 6). These researchers stated that the use of biologics for guttate psoriasis is not standard of care, as it is commonly self-limited, and biologics are intended for long-term control of chronic plaque psoriasis. However, a phase-IIIb trial of patients with chronic plaque psoriasis comparing standard ustekinumab maintenance dosing (45 or 90 mg every 12 weeks) with extended tailored dosing (16, 20, and 24 week intervals) showed that some patients with chronic plaque psoriasis can remain clear/almost clear. They noted that it is possible that after induction with ustekinumab, select patients with chronic plaque or chronic guttate psoriasis may obtain disease control with ustekinumab induction or intermittent dosing. The authors stated that to-date, there is only 1 published case report of a chronic guttate psoriasis patient who was successfully treated with ustekinumab. The authors presented 6 cases of recalcitrant guttate psoriasis successfully treated with ustekinumab. However, they acknowledged the drawbacks of this series: small sample size (n = 6), lack of control group, and variable ustekinumab dosing used. This was not a prospective clinical trial, and the researchers recognized the need for further clinical and laboratory studies to help elucidate the molecular mechanisms driving the pathogenesis of guttate psoriasis. They noted that this limited case-series study provided evidence for the potential benefit of ustekinumab in the treatment of recalcitrant or chronic guttate psoriasis.

Hidradenitis Suppurativa and Hidradenitis Suppurativa Sarcoidosis

Lee and Eisen (2015) stated that given the absence of significant improvement in the treatment of hidradenitis suppurativa (HS) with traditional medical and surgical therapies, biologics have piqued the interest of research investigators. The effectiveness of biologics in the treatment of inflammatory conditions like psoriasis and rheumatoid arthritis (RA) is well-documented. More recently, success with biologics has been demonstrated in atopic dermatitis, another dermatological condition associated with inflammatory states. Researchers have begun to probe the utility of biologic agents in less prevalent conditions that feature inflammation as a key characteristic, namely, HS. Five agents in particular adalimumab, anakinra, etanercept, infliximab, and ustekinumab, have been explored in the setting of HS. The authors noted that results to-date put forward adalimumab and infliximab as biologic treatments that can safely be initiated with some expectant efficacy; other biologic agents require more rigorous examination before they can be added to the treatment armamentarium.

In a multi-center case-series study and systematic review, Montero-Vilchez and colleagues (2022) retrospectively examined the therapeutic outcomes of ustekinumab of patients with HS, and evaluated all published scientific evidence on its utilization in patients with HS. These researchers examined the therapeutic outcomes of 10 patients with HS treated with ustekinumab and conducted a systematic review of published epidemiological studies on ustekinumab-treated patients with HS. In the case-series study, an improvement in the Physician Global Assessment score was observed in 70 % (7/10) patients and an improvement in the Numerical Pain Rating Scale in 80 % (8/10). In the systematic review, clinical improvement in disease severity was reported in 76 % (34/45) patients and symptomatic improvement in 84 % (38/45). No severe ustekinumab-related AE was recorded. The authors concluded that these findings suggested that ustekinumab may be a safe and effective therapeutic option for patients with HS who failed to respond to 1st-line therapies.

Romani and associates (2020) stated that adalimumab is the only approved biological therapy for HS. The last published recommendations support the use of other off-label biologic therapies. These investigators reported on a multi-centric, retrospective review of patients with HS treated with an ustekinumab dosing schedule of intravenous infusion adjusted by weight, followed by a subcutaneous maintenance dose of 90-mg every 8 weeks, as recently approved for Crohn's disease. The minimal follow-up period required for inclusion was 16 weeks. A total of 14 patients from 6 hospitals were included. In 50 % of the treated patients, therapeutic outcomes, measured by means of the Hidradenitis Suppurativa Clinical Response (HiSCR) and decrease of Dermatology Life Quality Index (DLQI) and visual analog scale (VAS) of pain, were reached at week 16. In 71.42 % of patients DLQI and VAS of pain improved, irrespective of achievement of HiSCR; 2 patients abandoned treatment due to lack of efficacy or patient preferences. No ustekinumab-related adverse effects were reported. The results were limited by the retrospective nature of the study, the short follow-up period (16 weeks), and the small patient number (n =14). This therapeutic regime proved to be safe and showed moderate efficacy in treating HS with failure to previous biologic therapy. The authors concluded that the efficacy of ustekinumab in HS should be tested in randomized controlled trials (RCTs).

Montero-Vilchez et al (2022) stated that HS is a chronic auto-inflammatory skin disease. Adalimumab is the only biologic agent available to treat HS; however, a lack of response is observed in some patients. Ustekinumab may be useful to treat patients with HS who did not respond to adalimumab. In a retrospective, case-series, multi-center study, these researchers examined the therapeutic outcomes of ustekinumab in a series of patients with HS and evaluated all published scientific evidence on its use in patients with HS. They examined the therapeutic outcomes of 10 patients with HS treated with ustekinumab and carried out a systematic review of published epidemiological studies on ustekinumab-treated patients with HS. In this case-series study, an improvement in the Physician Global Assessment score was observed in 70 % (7/10) patients and an improvement in the Numerical Pain Rating Scale in 80 % (8/10). In the systematic review, clinical improvement in disease severity was reported in 76 % (34/45) patients and symptomatic improvement in 84 % (38/45). No severe ustekinumab-related AE was recorded. The authors concluded that these findings suggested that ustekinumab may be a safe and effective therapeutic option for patients with HS who failed to respond to 1st-line therapies. Moreover, these researchers stated that the drawbacks of this study included the small number of patients (n = 10), their heterogeneity, the retrospective study design, and differences in the outcome evaluation instruments and follow-up periods considered.

Large Vessel Vasculitis and Polymyalgia Rheumatic

Hellmich (2016) noted that imaging methods, such as joint and color duplex sonography, magnetic resonance imaging (MRI) and positron emission tomography (PET) nowadays facilitate the diagnosis of polymyalgia rheumatica and large vessel vasculitis (LVV) and have now been included in the new classification criteria. In patients with typical symptoms, color duplex sonography of the temporal artery can replace a biopsy of the temporal artery for the diagnosis of giant cell arteritis (GCA); however, the role of these methods for patient follow-up and assessment of prognosis is unclear. Polymyalgia rheumatica is treated with glucocorticoids (GC) in an initial dosage of up to 20 mg/day. In patients with LVV higher doses are needed for induction of remission. Furthermore, the rate of relapse and GC-related AEs are higher in GCA and Takayasu arteritis (TA). Thus, initial GC-sparing treatment with methotrexate or other immunosuppressants is recommended. Recent studies showed an effectiveness of biologics. Recent data of the placebo-controlled proof of concept trials showed that the IL-6 antagonist tocilizumab reduced GC requirements and relapse rates in patients with GCA and polymyalgia rheumatica. Both ustekinumab and abatacept appeared to be effective in recent pilot trials for GCA. Antibodies against TNF alpha were ineffective for polymyalgia rheumatica and GCA in placebo-controlled trials; but data from open label studies suggested some effectiveness in refractory TA.

Koster and colleagues (2016) reviewed advances in the management of GCA and TA focusing on recent developments in targeted biologic therapy. The role of biologics in the treatment of LVV is expanding; TNF-alpha inhibitors appear to be effective in the treatment of TA but have little benefit in GCA. Preliminary clinical trial data suggested that abatacept and tocilizumab reduce the risk of relapse in GCA. Increasing observational evidence supported the use of IL-6 inhibitors in TA. Based on a small, open-label study, ustekinumab appeared safe and potentially effective for refractory GCA. A possible role of B cell dysregulation may contribute to pathogenic mechanisms in LVV, but support for the use of B cell depleting therapy is limited. The authors concluded that IL-6 inhibitors appeared effective in the treatment of refractory cases of LVV; however, utility in newly diagnosed immunosuppressive-naïve patients is less well established. They stated that abatacept and ustekinumab are promising targets for therapy in LVV; but further investigation is needed before routine use is considered.

Roberts and Clifford (2017) stated that GCA is a large vessel vasculitis that may be associated with significant complications such as blindness, stroke, or aortic aneurysm and dissection in a subset of patients. Given the serious side effects associated with prolonged courses of glucocorticoids and frequent relapses experienced when doses are tapered, increased efforts are being dedicated to the discovery of safer and more effective therapies to control this disease. These investigators examined the role of glucocorticoid-sparing agents in the medical management of GCA with a special focus on the most recent evidence regarding the role of promising biologic agents (e.g., abatacept, tocilizumab, and ustekinumab) and other novel therapies.

Keser et al (2022) stated that LVV, including Takayasu arteritis (TAK) and GCA, causes granulomatous vascular inflammation mainly in large vessels, and is the most common primary vasculitis in adults. Vascular inflammation may evoke many clinical features including vision impairment, stroke, limb ischemia, and aortic aneurysms. The best way to diagnose LVV is to combine medical history, physical examination, various laboratory tests, and imaging modalities. Progress in imaging modalities facilitated early diagnosis and follow-up of the disease activity. Conventional angiography is no longer the gold standard for the diagnosis of TAK. Similarly, temporal artery biopsy is no longer the only tool for diagnosing cranial GCA. In selected cases, color Doppler ultrasound may be used for this purpose. Despite some similarities, TAK and GCA differ in many aspects and they are different diseases. They also have different clinical subtypes. The presence of aortitis does not always implicate the diagnosis of TAK or GCA; infectious aortitis, as well as non-infectious aortitis associated with other autoimmune rheumatic diseases should be excluded. Treatment of LVV includes glucocorticoids (GCs), conventional immunosuppressive agents, and biological drugs; TNF inhibitors are ineffective in GCA but effective in TAK. On the other hand, tocilizumab may be used to treat both diseases. Promising targeted therapies evaluated in ongoing clinical trials include ustekinumab, secukinumab, anakinra, guselkumab, abatacept, tofacitinib, upadacitinib, mavrilimumab, and bosentan.

Leukocyte Adhesion Deficiency Type 1

Moutsopoulos and co-workers (2017) reported on the case of a patient with leukocyte adhesion deficiency type 1 (LAD1) who had severe periodontitis and an intractable, deep, non-healing sacral wound. These researchers had previously found a dominant IL-23-IL-17 signature at inflamed sites in humans with LAD1 and in mouse models of the disorder. Blockade of this pathway in mouse models had resulted in resolution of the immunopathologic condition. These investigators treated the patient with ustekinumab, an antibody that binds the p40 subunit of IL-23 and IL-12 and thereby blocks the activity of these cytokines, inhibiting IL-23-dependent production of IL-17. After 1 year of therapy, the patient had resolution of his inflammatory lesions without serious infections or adverse reactions. Inhibition of IL-23 and IL-17 may have a role in the management of LAD1. The authors stated that the dosing of ustekinumab that is normally used for psoriasis was effective in early complete blocking of IL-17 and related cytokines and chemokines in the gingiva for at least 3 weeks. Whether a shorter treatment interval would have more clinical benefit is unclear. Moreover, they stated that it is important to recall that skin-wound closure and complete epithelialization minimizes microbial stimulation; in contrast, the periodontal pocket is an open environment that is continuously subjected to a high level of microbial stimulation. Thus, although a limited course of this targeted therapy might help in the treatment of a closable wound, LAD1 periodontitis is an ongoing process that is likely to require more long-term therapy. The authors stated that treatment of additional patients with LAD1 is needed to help clarify the role of ustekinumab in the management of immunopathologic processes in LAD1 and the long-term safety of the drug in light of the underlying immunodeficiency in these patients.

Microscopic Colitis

In a systematic review and meta-analysis, Taneja et al (2022) examined the safety and effectiveness of biologic therapy in the treatment of steroid-refractory microscopic colitis (MC). These investigators searched Medline, Embase, Web of Science, and Cochrane Central to identify studies and abstracts reporting safety or effectiveness data on biologic use (infliximab, adalimumab, certolizumab, golimumab, vedolizumab, ustekinumab, and tofacitinib) for induction and maintenance of remission in MC. They evaluated clinical remission and response rates and all reported AEs. A total of 376 studies were screened yielding 13 articles (including 4 abstracts) with a combined information on 78 patients for safety and effectiveness outcomes. Most studies were case series. Vedolizumab was used in 5 studies, adalimumab in 3, and a combination of infliximab and adalimumab in 5 studies. The rates of remission were 66.08 % (95 % CI: 36.79 % to 95.37 %; I² = 71 %) at weeks 3 to 6 and 54.20 % (95 % CI: 39.39 % to 69.01 %; I² = 0 %) at weeks 12 to 16. Clinical response rates were 100 % (95 % CI: 88.04 % to 100 %; I² = 0 %) at weeks 3 to 6 and 67.20 % (95 % CI: 47.72 % to 86.69 %; I² = 52 %) at weeks 12 to 16. Most frequent AE was medication discontinuation with a pooled incidence of 16.1 % (95 % CI: 5.9 % to 37.5 %). No deaths attributable to biologic use were reported. The overall quality of evidence was very low due to the high risk of biases. The authors concluded that low-quality evidence supports the short-term effectiveness of biologics in budesonide-refractory MC. Moreover, these researchers stated that while these findings represent the most comprehensive evaluation of biologic therapy in severe MC, further research including randomized clinical trials is needed to better define the role of specific agents and long-term effectiveness.

Furthermore, the guidelines issued by the United European Gastroenterology and European Microscopic Colitis Group do not provide a recommendation for the use of ustekinumab for the treatment of microscopic colitis.

Other Dermatological Conditions

Rawal et al (2022) stated that ustekinumab is approved for the treatment of psoriasis and CD. Because many dermatological conditions are due to immune-mediated development, ustekinumab may be effective in other conditions. In a systematic review of the off-label uses of ustekinumab, as well as on-label adverse effect, these investigators reported on clinical improvement. Medline, Embase, Web of Science, and Cochrane databases were searched for studies regarding ustekinumab treatment of hidradenitis suppurativa (HS), lichen planus (LP), pyoderma gangrenosum (PG), pityriasis rubra pilaris (PRP), alopecia areata (AA), atopic dermatitis (AD), Bechet's disease, bullous pemphigoid (BP), hidradenitis suppurativa, sarcoidosis, cutaneous systemic lupus erythematosus (SLE), and vitiligo. Descriptive statistics were carried out. A total of 74 articles of 4,596 screened were included; and reported on 212 patients receiving ustekinumab treatment. Across all studies, ustekinumab showed promise in treating patients: AA (10/12 patients; 83.3 % improvement), AD (28/74 patients; 37.8 % improvement), HS (42/52 patients; 80.8 % improvement), and PRP (25/27 patients; 92.6 % improvement), among others; AEs were noted with the use of ustekinumab, including development of AA (4 patients), AD (3 patients), and BP (4 patients), among others. The authors concluded that ustekinumab can be a promising therapeutic option for patients with dermatological conditions refractory to traditional therapies; AEs must be monitored in certain patients.

Wu and Dai (2022) noted that IL-17 inhibitors, including secukinumab, brodalumab, and ixekizumab, have been FDA-approved for the treatment of psoriasis. In addition to psoriasis, IL-17 has been implicated in the pathophysiology of other inflammatory skin conditions. These investigators examined the evidence on off-label dermatologic uses of IL-17 inhibitors. They carried out searches in PubMed and ClinicalTrials.gov for clinical trials, observational studies, case series, and case reports evaluating non-psoriatic uses of the 3 IL-17 inhibitors. Studies examined the effectiveness of IL-17 inhibitors for the following conditions: HS, PRP, Behcet's disease, AA, and allergic contact dermatitis. Based on the available literature, secukinumab appeared to be a potential treatment for HS, PRP, and Behcet's disease, while ixekizumab appeared to be a potential treatment for HS and PRP. Moreover, the authors concluded that more clinical trials data are needed to examine the safety and effectiveness of IL-17 inhibitors for the treatment of these conditions.

Ye et al (2022) stated that ustekinumab is a human IL-12 and IL-23 antagonist and has been indicated for the treatments of moderate-to-severe plaque psoriasis, psoriatic arthritis, CD and UC. These investigators examined the evidence on off-label uses of ustekinumab. They conducted searches in PubMed and ClinicalTrials.gov for clinical trials, observational studies, case series, and case reports evaluating label uses of ustekinumab. Studies evaluated the effectiveness of ustekinumab for the following conditions: other types of psoriasis (except plaque psoriasis and psoriatic arthritis), PRP, HS, AD, PG, etc. The authors concluded that based on the available literature, ustekinumab appeared to be a potential treatment choice for many other diseases. Moreover, these investigators stated that more clinical trials data are needed to examine the safety and effectiveness of ustekinumab for the treatment of these conditions.

Pityriasis Rubra Pilaris

Lwin and colleagues (2018) noted that pityriasis rubra pilaris (PRP) represents a group of rare chronic inflammatory skin disorders in which approximately 1 in 20 affected individuals show autosomal dominant inheritance. In such cases, there may be gain-of-function mutations in CARD14, encoding caspase recruitment domain-containing protein 14 (CARD14) that activates the non-canonical nuclear factor-kappa B (NF-κB) pathway, thereby promoting cutaneous inflammation. These investigators reported on the cases of a mother and son with PRP due to a new missense mutation in CARD14 and described the beneficial clinical effects of ustekinumab in both subjects. A 49-year-old woman and her 20-year-old son had lifelong, generalized, patchy

erythematous scale with a few islands of sparing, as well as minor nail ridging and mild palmoplantar keratoderma, features consistent with generalized PRP. Topical steroids, phototherapy and oral retinoids proved ineffective therapies. Following informed consent, Sanger sequencing of CARD14 in both individuals revealed a new heterozygous single nucleotide transversion in exon 4, c.356T>G, resulting in the missense mutation, p.Met119Arg. The authors reported that ustekinumab, at a dose of 45-mg every 12 weeks, brought about a significant physical and emotional improvement in both the mother and son within a few days of the initial dose, which was sustained on maintenance dosing. This report highlighted the therapeutic potential of biologics that down-regulate NF- κ B signaling in familial PRP with mutations in CARD14. These preliminary findings need to be validated in well-designed studies.

Plaque Psoriasis and Psoriatic Arthritis

In September 2009, the FDA approved ustekinumab (Stelara) for the treatment of adults with moderate-to-severe psoriasis. Ustekinumab suppresses IL-12- and IL-23-mediated inflammation associated with psoriasis. In a review on the use of biological agents in the treatment of psoriasis, Tzu et al (2008) noted that IL-12/IL-23 inhibitors are new agents. Furthermore, Rozenblit and Lebwohl (2009) stated that new biologic therapies for psoriasis and psoriatic arthritis include antibodies to IL-12 and IL-23.

Weber and Kearn (2009) noted that in 2 large, phase III trials in patients with moderate-to-severe plaque psoriasis, significantly more subcutaneous ustekinumab 45 or 90 mg recipients (administered as 2 injections 4 weeks apart) than placebo recipients achieved a 75 % improvement on the PASI 75 score at 12 weeks. Other efficacy measures, including the physician's global assessment (PGA) of clinical response at week 12, also favored ustekinumab over placebo. Psoriatic symptom control was maintained during ustekinumab maintenance therapy (administered once every 12 weeks) for up to 76 weeks. In a phase II trial in patients with active plaque psoriasis and psoriatic arthritis, signs and symptoms of arthritis and psoriatic symptom control were improved to a greater extent with ustekinumab than with placebo at 12 weeks, based on the proportion of patients achieving a 20 % improvement in ACR response criteria (arthritis) or PASI 75 (skin symptoms). Health-related quality of life, assessed using the Dermatology Life Quality Index and the Health Assessment Questionnaire disability index, was improved to a significantly greater extent with ustekinumab than with placebo at week 12. Subcutaneous ustekinumab was generally well-tolerated in clinical trials, with most treatment-emergent adverse events being of mild severity.

In a phase III, parallel, randomized, double-blind, placebo-controlled trial (PHOENIX 1), Leonardi et al (2008) examined the safety and effectiveness of ustekinumab for the treatment of psoriasis. A total of 766 patients with moderate-to-severe psoriasis were randomly assigned to receive ustekinumab 45 mg (n = 255) or 90 mg (n = 256) at weeks 0 and 4 and then every 12 weeks; or placebo (n = 255) at weeks 0 and 4, with subsequent cross-over to ustekinumab at week 12. Patients who were initially randomized to receive ustekinumab at week 0 who achieved long-term response (at least 75 % improvement in psoriasis area and severity index [PASI 75] at weeks 28 and 40) were re-randomized at week 40 to maintenance ustekinumab or withdrawal from treatment until loss of response. Both randomizations were done with a minimization method via a centralized interactive voice response system. The primary endpoint was the proportion of patients achieving PASI 75 at week 12. Analyses were by intention-to-treat. All randomized patients were included in the efficacy analysis. Overall, 171 (67.1 %) patients receiving ustekinumab 45 mg, 170 (66.4 %) receiving ustekinumab 90 mg, and 8 (3.1 %) receiving placebo achieved PASI 75 at week 12 (difference in response rate versus placebo 63.9 %, 95 % confidence interval [CI]: 57.8 to 70.1, p < 0.0001 for 45 mg and 63.3 %, 57.1 to 69.4, p < 0.0001 for 90 mg). At week 40, long-term response had been achieved by 150 patients in the 45-mg group and 172 patients in the 90-mg group. Of these, 162 patients were randomly assigned to maintenance ustekinumab and 160 to withdrawal. PASI 75 response was better maintained to at least 1 year in those receiving maintenance ustekinumab than in those withdrawn from treatment at week 40 (p < 0.0001 by log-rank test). During the placebo-controlled phase, adverse events occurred in 278 (54.5 %) of the 510 patients receiving ustekinumab and 123 (48.2 %) of the 255 receiving placebo. Serious adverse events occurred in 6 (1.2 %) of 510 patients receiving ustekinumab and in 2 (0.8 %) of 255 receiving placebo in this phase. The pattern of adverse events was much the same in the placebo cross-over and randomized withdrawal phases as it was in the placebo-controlled phase. The authors concluded that ustekinumab seems to be effective for the treatment of moderate-to-severe psoriasis; dosing every 12 weeks maintains efficacy for at least 1 year in most patients.

In a multi-center, phase III, double-blind, placebo-controlled trial (PHOENIX 2), Papp et al (2008) assessed the safety and effectiveness of ustekinumab in psoriasis patients and assessed dosing intensification in partial responders. A total of 1,230 patients with moderate-to-severe psoriasis (defined by a PASI score greater than or equal to 12, and at least 10 % total body surface area involvement) were randomly assigned to receive ustekinumab 45 mg (n = 409) or 90 mg (n = 411) at weeks 0 and 4, then every 12 weeks, or placebo (n = 410). Partial responders (i.e., patients achieving greater than or equal to 50 % but less than 75 % improvement from baseline in PASI) were re-randomized at week 28 to continue dosing every 12 weeks or escalate to dosing every 8 weeks. Both randomizations were done with a minimization method via a centralized interactive voice response. The primary endpoint was the proportion of patients achieving at least PASI 75 at week 12. Analyses were by intention-to-treat. All randomized patients were included in the efficacy analysis. Overall, 273 (66.7 %) patients receiving ustekinumab 45 mg, 311 (75.7 %) receiving ustekinumab 90 mg, and 15 (3.7 %) receiving placebo achieved the primary endpoint (difference in response rate 63.1 %, 95 % CI: 58.2 to 68.0, p < 0.0001 for the 45-mg group versus placebo and 72.0 %, 67.5 to 76.5, p < 0.0001 for the 90-mg group versus placebo). More partial responders at week 28 who received ustekinumab 90 mg every 8 weeks achieved PASI 75 at week 52 than did those who continued to receive the same dose every 12 weeks (22 [68.8 %] versus 11 [33.3 %]; difference in response rate 35.4 %, 95 % CI: 12.7 to 58.1, p = 0.004). There was no such response to changes in dosing intensity in partial responders treated with ustekinumab 45 mg. During the placebo-controlled phase, 217 (53.1 %) patients in the 45-mg group, 197 (47.9 %) in the 90-mg group, and 204 (49.8 %) in the placebo group experienced adverse events; serious

adverse events were seen in 8 (2.0 %) patients in the 45-mg group, 5 (1.2 %) in the 90-mg group, and 8 (2.0 %) in the placebo group. The authors concluded that although treatment with ustekinumab every 12 weeks is effective for most patients with moderate-to-severe psoriasis, intensification of dosing to once every 8 weeks with ustekinumab 90 mg might be necessary to elicit a full response in patients who only partially respond to the initial regimen.

Landells et al (2015) stated safe and effective therapies are needed for pediatric patients with psoriasis. On October 13, 2017, Janssen Biotech, Inc., announced that the FDA approved an expanded indication for Stelara (ustekinumab) for the treatment of adolescents (12 years of age or older) with moderate to severe plaque psoriasis who are candidates for phototherapy or systemic therapy. The FDA-approval of ustekinumab in adolescents (12 years or older) was based on data from a multicenter, randomized, double blind, placebo-controlled study that evaluated ustekinumab in patients age 12 to 17 years who had moderate-to-severe psoriasis. Patients (n = 110) were randomly assigned to ustekinumab standard dosing (SD; 0.75 mg/kg [≤ 60 kg], 45 mg [$>60\text{--}\leq 100$ kg], and 90 mg [>100 kg]) or half-standard dosing (HSD; 0.375 mg/kg [≤ 60 kg], 22.5 mg [$>60\text{--}\leq 100$ kg], and 45 mg [>100 kg]) at weeks 0 and 4 and every 12 weeks or placebo at weeks 0 and 4 with crossover to ustekinumab SD or HSD at week 12. Clinical assessments included the proportion of patients achieving a Physician's Global Assessment of cleared/minimal (PGA 0/1), at least 75% improvement in Psoriasis Area and Severity Index (PASI 75), and at least 90% in PASI (PASI 90). Adverse events (AEs) were monitored through week 60. At week 12, 67.6% and 69.4% of patients receiving ustekinumab HSD and SD, respectively, achieved PGA 0/1 versus 5.4% for placebo ($P < .001$). Significantly greater proportions receiving ustekinumab achieved PASI 75 (HSD, 78.4%; SD, 80.6%; placebo, 10.8%) or PASI 90 (HSD, 54.1%; SD, 61.1%; placebo, 5.4%) at week 12 ($P < .001$). Through week 12, 56.8% of placebo patients, 51.4% of HSD patients, and 44.4% of SD patients reported at least one AE; through week 60, 81.8% reported AEs. The study was small relative to adult trials. The authors concluded that in this patient population (12-17 years), the standard ustekinumab dose provided response comparable to that in adults with no unexpected AEs through 1 year.

In a phase II, randomized, double-blind, placebo-controlled, cross-over study, Gottlieb et al (2009) evaluated the safety and effectiveness of ustekinumab for psoriatic arthritis. Patients with active psoriatic arthritis were randomly allocated via interactive voice response system to either ustekinumab (90 mg or 63 mg) every week for 4 weeks (weeks 0 to 3) followed by placebo at weeks 12 and 16 (n = 76; Group 1) or placebo (weeks 0 to 3) and ustekinumab (63 mg) at weeks 12 and 16 (n = 70; Group 2). The first 12 weeks of the study were placebo-controlled. Masking was maintained to week 16, and patients were followed-up to week 36 [corrected]. The primary endpoint was American College of Rheumatology 20 % improvement (ACR20) response at week 12. Analysis was by intention-to-treat. At week 12, 32 (42 %) patients in Group 1 and 10 (14 %) in Group 2 achieved the primary endpoint (difference 28 % [95 % CI: 14.0 to 41.6]; $p = 0.0002$). Of 124 (85 %) participants with psoriasis affecting 3 % or more body surface area, 33 of 63 (52 %) in Group 1 and 3 of 55 (5 %) in Group 2 had a 75 % or greater improvement in psoriasis area and severity index score at week 12 (47 % [33.2 to 60.6]; $p < 0.0001$). During the placebo-controlled period (weeks 0 to 12), adverse events arose in 46 (61 %) patients in Group 1 and 44 (63 %) in Group 2; serious adverse events were recorded in 3 (4 %) Group 2 patients (none in Group 1). The authors concluded that ustekinumab significantly reduced signs and symptoms of psoriatic arthritis and diminished skin lesions compared with placebo, and the drug was well-tolerated. They stated that larger and longer term studies are needed to further characterize ustekinumab safety and effectiveness for the treatment of psoriatic arthritis.

Ryan et al (2010) examined the safety and effectiveness of ustekinumab in all clinical studies to date, using PubMed listed publications and official product websites. Ustekinumab has shown significant effectiveness in the treatment of chronic plaque psoriasis in phase III studies, and promising results in phase II studies in psoriatic arthritis. Effectiveness has been shown in Crohn's disease only in non-responders to infliximab. Furthermore, ustekinumab did not show benefit in the treatment of multiple sclerosis.

The effectiveness of ustekinumab in treating psoriasis were reproduced in the phase III, multi-center, randomized ACCEPT trial (2008). This head-to-head study comparing ustekinumab and etanercept for the treatment of moderate-to-severe psoriasis showed ustekinumab superior to etanercept according to primary and major secondary efficacy endpoints. The primary endpoint of the trial was the percentage of participants achieving at least a 75 % reduction in psoriasis at week 12 as measured by PASI 75. At week 12, after 2 subcutaneous injections at weeks 0 and 4, 68 % and 74 % of patients receiving ustekinumab 45 mg or ustekinumab 90 mg, respectively, achieved a PASI 75 compared with 57 % of patients receiving etanercept 50 mg subcutaneous injections twice-weekly for 12 weeks ($p = 0.012$ for ustekinumab 45 mg; $p < 0.001$ for ustekinumab 90 mg, each compared with etanercept).

In a phase II clinical trial, Kavanaugh et al (2010) evaluated the effect of ustekinumab on physical disability and health-related quality of life (HRQoL) in patients with psoriatic arthritis (PsA). Patients with active PsA were randomized (1:1 ratio) to receive either ustekinumab at weeks 0, 1, 2, and 3 and placebo at weeks 12 and 16 (n = 76) or placebo at weeks 0, 1, 2, and 3 and ustekinumab at weeks 12 and 16 (n = 70). Physical function was assessed using the disability index from the Health Assessment Questionnaire-Disability Index (HAQ-DI) in all randomized patients. HRQoL was evaluated using the DLQI in a subset of patients (84.9 %) with at least 3 % body surface area (BSA) psoriasis involvement at baseline. At baseline, overall mean HAQ-DI and DLQI scores were 0.9 and 11.5, respectively, indicating impaired physical function and moderate effect on HRQoL. At week 12, ustekinumab patients had significantly more improvement (decrease) in the mean HAQ-DI (-0.31) and DLQI (-8.6) scores versus placebo (-0.04 and -0.8, respectively; $p < 0.001$ for both comparisons). At week 12, 58.7 % (37/63) of ustekinumab-treated patients had a DLQI score of 0 or 1 (no negative effect of disease or treatment on HRQoL) versus 5.5 % (3/55) for placebo ($p < 0.001$). The results also indicated a positive but weak correlation between improvement in physical function and HRQoL, pain, and skin response as well as between improvement in joint and skin responses in patients receiving

ustekinumab or placebo. The authors concluded that ustekinumab significantly improved physical function and HRQoL in patients with PsA and psoriasis involving at least 3 % BSA. Limitations of the study included the short duration of the placebo-controlled period and the relatively small patient population.

Kurzeja et al (2011) stated that interleukin (IL)-23 is an important regulator of T helper-17 lymphocytes, which influence the cutaneous immune system by production of IL-17 and several other pro-inflammatory cytokines. This pathway has been recently linked to the pathogenesis of psoriasis and numerous other skin diseases. A newly developed biologic drug, ustekinumab, which targets the p40 subunit of IL-12 and IL-23, was approved by the U.S. FDA and the European Medicines Agency in 2009 for the treatment of moderate to severe psoriasis. Administered as subcutaneous injections of 45 mg at weeks 0 and 4, and then every 12 weeks, ustekinumab produces a 75 % improvement in the PASI in 66.4 to 75.7 % of patients and a DLQI score of 0 or 1 in 55 to 56 % of patients after 12 weeks of therapy. A recent clinical trial also indicates the possible efficacy of ustekinumab in psoriatic arthritis. The proportion of patients who had at least 1 adverse event through 12 weeks in clinical studies was 51.6 to 57.6 % in the ustekinumab group and 50.4 % in the placebo group. Serious adverse events were observed in 1.4 to 1.6 % of patients treated with ustekinumab and in 1.4 % of patients receiving placebo. Injection-site reactions occurred in 1 to 2 % of patients and 5 % of patients developed anti-ustekinumab antibodies. The authors concluded that further studies are needed to evaluate the long-term safety and effectiveness of ustekinumab.

U.S. Food and Drug Administration (FDA) approved ustekinumab (Stelara), alone or in combination with methotrexate, for the treatment of adult patients (18 years or older) with active psoriatic arthritis. The approval was supported by findings from two pivotal phase 3 multicenter, randomized, double-blind, placebo-controlled trials of ustekinumab administered subcutaneously in subjects with active psoriatic arthritis (PSUMMIT I and PSUMMIT II), which evaluated the efficacy and safety of subcutaneously administered ustekinumab 45 mg or 90 mg at weeks 0, 4 and then every 12 weeks. The trials included 927 patients diagnosed with active psoriatic arthritis who had at least five tender and five swollen joints and C-reactive protein (CRP) levels of at least 0.3 mg/dL in spite of previous treatment with conventional therapy. PSUMMIT II also included 180 patients with previous exposure to 1-5 tumor necrosis factor (TNF) inhibitors. Results from PSUMMIT I showed that at week 24, 42 percent and 50 percent of patients receiving ustekinumab 45 mg and 90 mg, respectively, achieved at least 20 percent improvement in signs and symptoms according to the American College of Rheumatology criteria (ACR 20), the primary endpoint for both studies. In PSUMMIT II, 44 percent of patients receiving ustekinumab 45 mg and 44 percent of patients receiving ustekinumab 90 mg achieved ACR 20 at week 24. Additionally, ustekinumab improved soft tissue components of the disease, including dactylitis (inflammation of the finger or toe), enthesitis (inflammation of the entheses, the sites where tendons or ligaments attach to bone) and skin component as measured by Psoriasis Area and Severity Index score (PASI) 75.

In August 2022, the U.S. FDA approved ustekinumab (Stelara) for the treatment of pediatric patients 6 years of age and older with active psoriatic arthritis (PsA). FDA approval was based on pharmacokinetic (PK) data and extrapolation of the established efficacy and existing safety profile of ustekinumab in multiple Phase 3 studies in adult and pediatric patients with moderate-to-severe plaque psoriasis (PsO) (PSTELLAR, CADMUS, and CADMUS Jr) and adult patients with active PsA (PSUMMIT I and II). "With the limited availability of pediatric PsA patients for inclusion in clinical trials, researchers utilized an extrapolation approach based on previous PK, efficacy and safety observations from a closely adjacent population of pediatric patients with moderate to severe plaque PsO who also had active PsA, as well as adult patients with moderate to severe plaque PsO or active PsA. An analysis of the data demonstrated that PK exposure of STELARA in these pediatric PsO patients with active PsA was consistent with that of Phase 3 clinical trials of STELARA in pediatric PsO patients without active PsA, as well as with adult patients with moderate to severe plaque PsO or adult patients with active PsA, while data on common efficacy endpoints were similar in these pediatric PsO patients with active PsA" (Janssen, 2022).

Pyoderma Gangrenosum

Guenova et al (2011) noted that IL-23 is involved in the pathogenesis of the chronic inflammatory Crohn disease. Pyoderma gangrenosum (PG) is often associated with and can even be the first manifestation of this disease and has abundant neutrophilic infiltration. Because IL-23 plays a critical role in driving inflammation associated with IL-17 production and especially neutrophil recruitment, these researches suspect that PG might be driven by a pathogenetic mechanism similar to that of inflammatory bowel diseases or psoriasis. Tissue sample analysis showed highly elevated expression of IL-23 on both transcriptional and protein level in a recalcitrant PG lesion. On the basis on these data, a treatment targeting the p40 subunit of the heterodimeric IL-23 with the monoclonal antibody ustekinumab was started. Therapy with ustekinumab resulted in a significant decrease of IL-23 expression in PG and healing after 14 weeks of treatment. No relapse occurred in a 6-month follow-up period. The authors concluded that these findings provided evidence of an IL-23-dominated inflammatory infiltrate in PG. This might specify a new concept for PG pathophysiology and suggests a possibility for using ustekinumab as a therapeutic agent in this disease.

Rheumatoid Arthritis

In a randomized phase-II clinical trial, Smolen and colleagues (2017) evaluated the safety and effectiveness of subcutaneously administered ustekinumab and guselkumab in patients with active rheumatoid arthritis (RA) despite methotrexate (MTX) therapy. Patients were randomly assigned (1:1:1:1:1) to receive placebo at weeks 0, 4 and every 8 weeks (n = 55), ustekinumab 90-mg at weeks 0, 4 and every 8 weeks (n = 55), ustekinumab 90-mg at weeks 0, 4 and every 12 weeks (n = 55), guselkumab 50-mg at weeks 0, 4 and every 8 weeks (n = 55), or guselkumab 200-mg at weeks 0, 4 and every 8 weeks (n = 54) through week 28; all patients continued a stable dose of MTX (10 to 25 mg/week). The primary end-point was the proportion of patients with at

least a 20 % improvement in the American College of Rheumatology criteria (ACR 20) at week 28. Safety was monitored through week 48. At week 28, there were no statistically significant differences in the proportions of patients achieving an ACR 20 response between the combined ustekinumab group (53.6 %) or the combined guselkumab group (41.3 %) compared with placebo (40.0 %) ($p = 0.101$ and $p = 0.877$, respectively). Through week 48, the proportions of patients with at least 1 AE were comparable among the treatment groups. Infections were the most common type of AE. The authors concluded that treatment with ustekinumab or guselkumab did not significantly reduce the signs and symptoms of RA.

SAPHO Syndrome

Firinu and colleagues (2016) stated that SAPHO syndrome (synovitis, acne, pustulosis, hyperostosis, and osteitis) is a rare autoimmune disease which, due to its clinical presentation and symptoms, is often mis-diagnosed and unrecognized. Its main features are prominent inflammatory cutaneous and articular manifestations. Treatments with immunosuppressive drugs have been used for the management of SAPHO with variable results. To-date, the use of anti-TNF-alpha agents has proved to be an effective alternative to conventional treatment for unresponsive or refractory SAPHO cases. TNF-alpha is a pro-inflammatory cytokine and pivotal regulator of other cytokines, including IL-1 β , IL-6, and IL-8, involved in inflammation, acute-phase response induction, and chemotaxis. IL-1 inhibition strategies with anakinra have shown efficacy as 1st and 2nd lines of treatment. These investigators described the main characteristics of biological drugs currently used for SAPHO syndrome. They also described some of the promising therapeutic effects of ustekinumab after failure of multiple drugs including anti-TNF-alpha and anakinra.

Wendling and associates (2017) noted that treatment of SAPHO syndrome is not standardized, and in case of inadequate response to anti-inflammatory drugs, the use of anti-TNF or anti-IL-1 biologic treatments has been reported. The IL-23/Th17 axis may be involved in SAPHO syndrome. These investigators reported the findings of 6 courses of IL-23 and IL-17 targeted therapies (3 ustekinumab and 3 secukinumab) in patients with SAPHO syndrome unresponsive to previous treatments (conventional synthetic disease-modifying anti-rheumatic drugs [csDMARDs] and biological DMARDs [bDMARDs]). With a mean treatment duration of 5.5 months, improvement of skin symptoms was noticed in 3 cases, 1 improvement with secukinumab and 2 remissions (1 with secukinumab, 1 with ustekinumab). Regarding the rheumatic symptoms, no major improvement was observed under any of the 6 treatment courses. No particular safety concerns were reported, except cases of paradoxical psoriasis flare in 1 under ustekinumab and the other case under secukinumab. The effectiveness of ustekinumab in the treatment of SAPHO syndrome needs to be validated in well-designed studies.

Sarcoidosis

Monast and co-workers (2017) stated that the molecular basis of sarcoidosis phenotype heterogeneity and its relationship to effective treatment of sarcoidosis have not been elucidated. Peripheral samples from sarcoidosis subjects who participated in a phase-II clinical trial of golimumab and ustekinumab were used to measure the whole blood transcriptome and levels of serum proteins. Differential gene and protein expression analyses were used to explore the molecular differences between sarcoidosis phenotypes as defined by extent of organ involvement. The same data were also used in conjunction with an enrichment algorithm to identify gene expression changes associated with treatment with study drugs compared to placebo. These analyses revealed marked heterogeneity among the 3 sarcoidosis phenotypes included in the study cohort, including striking differences in enrichment of the interferon pathway. Conversely, enrichments of multiple pathways, including T cell receptor signaling, were similar among phenotypes. These investigators also identified differences between treatment with golimumab and ustekinumab that may explain the differences in trends for clinical efficacy observed in the trial. They found that molecular heterogeneity is associated with sarcoidosis in a manner that may be related to the extent of organ involvement. They stated that these findings may help to explain the difficulty in identifying clinically effective sarcoidosis treatments and suggested hypotheses for improved therapeutic strategies.

Spondyloarthropathy

Her and Kavanaugh (2013) noted that inhibitors of TNF have demonstrated dramatic clinical efficacy in patients with spondyloarthropathy (SpA). However, not all patients respond, and some patients who initially improve, subsequently lose response. Therefore, there is still an unmet clinical need for additional therapies. These researchers described the recent data on newer treatments for SpA patients. Treatments targeting various cytokines, cell surface molecules, and signaling molecules have been assessed. The effects of targeting B cells with rituximab, T-cell co-stimulation with abatacept, and interleukin (IL)-6 with tocilizumab have been disappointing in ankylosing spondylitis (AS). Abatacept appears to have a modest effect in patients with psoriatic arthritis (PsA). Targeting IL-17 with secukinumab, IL-12/23 with ustekinumab, and phosphodiesterase 4 (PDE4) with apremilast may prove to be promising treatments for SpA. The authors concluded that there are several newer therapies that may emerge for SpA, particularly those targeting IL-17, IL-23/IL-12, and PDE4.

Systemic Lupus Erythematosus

In a randomized, placebo-controlled, phase-III clinical trial, van Vollenhoven et al (2022) examined the safety and effectiveness of ustekinumab in the treatment of patients with active SLE despite receiving standard-of-care (SOC). Active SLE patients (SLE Disease Activity Index 2000 (SLEDAI-2K) of 6 or higher during screening and SLEDAI-2K of 4 or higher for clinical features at week 0) despite receiving oral glucocorticoids, anti-malarials, or immunomodulatory drugs were randomized (3:2) to receive

ustekinumab (intravenous infusion approximately 6 mg/kg at week 0, followed by subcutaneous injections of ustekinumab 90 mg at week 8 and every 8 weeks) or placebo through week 48. The primary endpoint was SLE Responder Index (SRI)-4 at week 52, and major secondary endpoints included time to flare through week 52 and SRI-4 at week 24. At baseline, 516 patients were randomized to placebo (n = 208) or ustekinumab (n = 308). Following the planned interim analysis, the sponsor discontinued the study due to lack of effectiveness but no safety concerns. Effectiveness analyses included 289 patients (placebo, n = 116; ustekinumab, n = 173) who completed or would have had a week 52 visit at study discontinuation. At week 52, 44% of ustekinumab patients and 56 % of placebo patients had an SRI-4 response; there were no appreciable differences between the treatment groups in the major secondary endpoints. Through week 52, 28 % of ustekinumab patients and 32 % of placebo patients had a British Isles Lupus Assessment Group flare, with a mean time to 1st flare of 204.7 and 200.4 days, respectively. Through week 52, 70 % of ustekinumab patients and 74 % of placebo patients had 1 or more AE(s). The authors concluded that ustekinumab did not show superiority over placebo in this population of adults with active SLE; AEs were consistent with the known safety profile of ustekinumab.

Ulcerative Colitis

Ulcerative colitis (UC) is a serious, chronic and progressive immune-mediated inflammatory disease of the large intestine, affecting approximately 910,000 people in the United States. Stelara (ustekinumab) targets the interleukin (IL)-12 and IL-23 cytokines which have been shown to play a role in inflammatory and immune responses (Janssen Pharmaceutical, 2019). The efficacy of ustekinumab was evaluated in a pivotal study for the induction and maintenance therapy in patients with UC.

Ustekinumab was evaluated as 8-week induction therapy and 44-week maintenance therapy in two randomized, double-blind, placebo-controlled clinical studies [UC-1 and UC-2 (NCT02407236)] in 961 adult patients with moderately to severely active ulcerative colitis who had an inadequate response to or failed to tolerate a biologic (i.e., TNF blocker and/or vedolizumab), corticosteroids, and/or 6-MP or AZA therapy. The 8-week intravenous induction study (UC-1) was followed by the 44-week subcutaneous randomized withdrawal maintenance study (UC-2) for a total of 52 weeks of therapy (Janssen Biotech, 2019). Sands et al. (2019) randomly assigned a total of 961 patients to receive an intravenous induction dose of ustekinumab (either 130 mg [320 patients] or a weight-range-based dose that approximated 6 mg per kilogram of body weight [322]) or placebo (319). Patients who had a response to induction therapy 8 weeks after administration of intravenous ustekinumab were randomly assigned again to receive subcutaneous maintenance injections of 90 mg of ustekinumab (either every 12 weeks [172 patients] or every 8 weeks [176]) or placebo (175). The primary end point in the induction trial (week 8) and the maintenance trial (week 44) was clinical remission (defined as a total score of ≤ 2 on the Mayo scale [range 0 to 12, with higher scores indicating more severe disease] and no subscore >1 [range, 0 to 3] on any of the four Mayo scale components). The authors found that the percentage of patients who had clinical remission at week 8 among those who received intravenous ustekinumab at a dose of 130 mg (15.6%) or 6 mg per kilogram (15.5%) was significantly higher than among patients who received placebo (5.3%) ($p < 0.001$ for both comparisons). Among patients who had a response to induction therapy with ustekinumab and underwent a second randomization, the percentage of patients who had clinical remission at week 44 was significantly higher among patients assigned to 90 mg of subcutaneous ustekinumab every 12 weeks (38.4%) or every 8 weeks (43.8%) than among those assigned to placebo (24.0%) ($p = 0.002$ and $p < 0.001$, respectively). The incidence of serious adverse events with ustekinumab was similar to that with placebo. Through 52 weeks of exposure, there were two deaths (one each from acute respiratory distress syndrome and hemorrhage from esophageal varices) and seven cases of cancer (one each of prostate, colon, renal papillary, and rectal cancer and three nonmelanoma skin cancers) among 825 patients who received ustekinumab, and no deaths and one case of cancer (testicular cancer) among 319 patients who received placebo. The authors concluded that ustekinumab was more effective than placebo for inducing and maintaining remission in patients with moderate-to-severe ulcerative colitis.

On October 21, 2019, the FDA approved Stelara (ustekinumab) for the treatment of adult patients with moderately to severely active ulcerative colitis. FDA approval was based on the pivotal Phase 3 UNIFI clinical trial [NCT02407236] which met its primary endpoint of clinical remission (Janssen Pharmaceutical, 2019).

Uveitis

Mugheddu and colleagues (2017) reported the first successful treatment of non-infectious uveitis with ustekinumab in a patient with severe concomitant psoriasis and psoriatic arthritis who failed to respond to conventional immune suppressants and with contraindications to TNF alpha inhibitors. The effectiveness of ustekinumab in the treatment of non-infectious uveitis needs to be further investigated.

Sota and associates (2018) stated that intra-ocular inflammation is one of the more relevant complications of Behcet's disease (BD), which tends to respond poorly to different medications. The ocular histopathologic changes are basically identical to those occurring in other organs and consist in a necrotizing leukocytoclastic obliterative vasculitis, which is probably immune complex-mediated and affects both arteries and veins of all sizes. There are growing evidences showing the potential role of biologic agents other than TNF-alpha agents in the management of ocular-BD, which have been collected in this review, including IL-1 and IL-6 blockade, secukinumab, ustekinumab, daclizumab, rituximab, and alemtuzumab. The authors concluded that further large studies are needed to fully elucidate and establish the effectiveness of these different agents in the refractory ocular manifestations of BD.

Furthermore, an UpToDate review on “Uveitis: Treatment” (Rosenbaum, 2017) does not mention ustekinumab as a therapeutic option.

Other Indications

Combined Ustekinumab and Vedolizumab for Inflammatory Bowel Disease

Ribaldone and colleagues (2019) noted that inflammatory bowel diseases (IBD) patients eligible for biological therapy represent a group with considerable disease burden and biologics only achieve 40 % clinical remission rates in responders after 1 year of therapy. These researchers collected all the published data regarding patients treated with dual biological therapy with an anti-TNF, ustekinumab or vedolizumab, for a period of at least 3 months and pooled the data regarding the safety and effectiveness. They carried out a Medline, and Web of Science search of all studies published in English until January 1, 2019. These investigators included 7 studies with a total of 18 patients; 15 patients were treated with a combination of an anti-TNF and vedolizumab, 3 patients were treated with vedolizumab and ustekinumab; 56 % of patients were affected by CD and 50 % of patients were treated with an immunosuppressant drug or steroid too. A clinical improvement was obtained in 100 % of patients, and an endoscopic improvement in 93 % of patients. No serious AEs were reported. The authors concluded that the use of dual biological therapy is an attractive therapeutic option and may be an opportunity to better tailor and personalize the therapies for patients. Moreover, these researchers stated that further studies, as randomized control trials (RCTs), to provide comparative safety and efficacy endpoints of combination therapies, and to clarify potential advantages of combined biological therapies, are needed.

Measurement of Serum Levels of Ustekinumab and Antibodies to Ustekinumab

Hsu and colleagues (2014) stated that specific anti-drug antibodies (ADAs) against biological agents may be clinically significant and potentially alter a biological drug's treatment efficacy. In a systematic review, these researchers determined the prevalence of ADAs against infliximab (IFX), etanercept (ETN), adalimumab (AdA), and ustekinumab in patients with psoriasis; examined if ADAs were associated with changes in drug efficacy; and examined the use of concomitant methotrexate to prevent ADA formation. Through a systematic search using Medline and Embase from January 29, 1950 to March 29, 2013, these researchers identified 25 studies that met the inclusion criteria. Of 7,969 patients with psoriasis, 950 tested positive for ADAs. Antibodies against IFX, ETN, AdA, and ustekinumab were reported in 5.4 to 43.6 %, 0 to 18.3 %, 6 to 45 % and 3.8 to 6 % of patients, respectively. Anti-IFX antibodies were associated with lower serum IFX concentrations in 3 studies, and decreased treatment response in 5 studies. ADAs against ETN were non-neutralizing and not associated with any apparent effects on clinical response. Anti-AdA antibodies were associated with lower serum AdA concentrations in 3 of 5 studies, and reduced clinical efficacy in 4 studies; 2 of 6 studies reported that anti-ustekinumab antibodies were associated with lower PASI responses, and 3 ustekinumab studies noted that most of these antibodies were neutralizing. The authors concluded that although the use of concomitant methotrexate with biological agents to prevent ADA formation in other immune-mediated diseases is promising, their use in psoriasis is sparse. Moreover, they stated that ADA development remains a challenge with biological therapies and therefore should be considered in patients with psoriasis who experience diminished treatment response.

In a prospective, observational study, Chiu and associates (2015) examined the immunogenicity of ustekinumab and its clinical consequences in patients with psoriasis. This trial enrolled 76 patients with plaque psoriasis who were treated with ustekinumab for a minimum of 7 months. Blood samples were drawn just prior to scheduled ustekinumab injection during clinic visits. Levels of anti-ustekinumab antibody (AUA) and serum ustekinumab concentration were measured respectively by radioimmunoassays (RIAs) and enzyme-linked immunoassays (ELISA), respectively, and correlated to clinical data and Psoriasis Area and Severity Index (PASI). AUA was detected in 6.5 % of patients after a mean of 13 months of treatment. Patients with positive AUA had significantly lower serum ustekinumab concentrations (0.01 versus 0.2 mg/L, $p < 0.001$) and lower PASI 50 response than patients without AUA (0 % versus 69 %, $p = 0.004$). The percentage of AUA formation was comparable between patients who had failed previous AdA with or without anti-AdA antibodies (AAA) (14.3 % versus 12.5 %, $p = 1.00$). However, a higher proportion of switchers without AAA obtaining PASI50 (71.4 % versus 37.5 %) and PASI75 response (42.9 % versus 12.5 %) within 7 months of ustekinumab treatment than with AAA although this difference did not reach statistical significance. The authors concluded that these findings suggested that presence of AUA was significantly associated with treatment failure for ustekinumab, although limited by a small sample size. Also, determining the presence of ADA to antecedent TNF- α antagonists may assist in choosing an optimized subsequent treatment modality achieving treatment success. They stated that additional studies involving a larger sample size, longer follow-up, and serial sampling is needed to strengthen the conclusion; assessing the immunogenic status in psoriasis patients treated with ustekinumab may be a rational approach for further decision-making and might assist in choosing an optimized treatment modality for the individual patient.

This study had several drawbacks, which included the observational cohort design and inadequate statistical power (less than 50 %) from small sample size of patient with AUA. Furthermore, each patient's blood sample was collected once during a clinic visit, although the timing of the collection was not uniform across all samples.

The Prescribing Information does not mention measurement of serum ustekinumab levels or levels of antibodies to ustekinumab (Janssen Biotech, 2020).

It is also interesting to note that an UpToDate review on “Treatment of psoriasis in adults” (Feldman, 2019) states that “Further study is necessary to determine whether assessing serum levels of adalimumab during treatment will be useful for improving responses to therapy”. Well-designed studies are needed to expand the existing evidence base to evaluate if therapeutic drug monitoring (e.g., using information provided by the Prometheus Anser UST test), would lead to changes in therapeutic interventions or other changes in disease management that ultimately improve patient health outcomes over the long-term.

Appendix

Examples of Clinical Reasons to Avoid Pharmacologic Treatment with Methotrexate, Cyclosporine, Acitretin, or Leflunomide

1. Clinical diagnosis of alcohol use disorder, alcoholic liver disease or other chronic liver disease
2. Drug interaction
3. Risk of treatment-related toxicity
4. Pregnancy or currently planning pregnancy
5. Breastfeeding
6. Significant comorbidity prohibits use of systemic agents (e.g., liver or kidney disease, blood dyscrasias, uncontrolled hypertension)
7. Hypersensitivity
8. History of intolerance or adverse event

Table: Brands of Targeted Immune Modulators and FDA-approved Indications (not an all-inclusive list)

Brand Name	Generic Name	FDA Labeled Indications
Actemra	tocilizumab	Coronavirus Disease 2019 (COVID-19) in hospitalized adults Cytokine release syndrome (CRS) Giant cell arteritis Juvenile idiopathic arthritis Rheumatoid arthritis Systemic juvenile idiopathic arthritis Systemic sclerosis-associated interstitial lung disease (SSc-ILD)
Arcalyst	rilonacept	Cryopyrin-associated periodic syndromes Deficiency of interleukin-1 receptor antagonist (DIRA) Recurrent pericarditis
Cimzia	certolizumab	Ankylosing spondylitis or non-radiographic axial spondyloarthritis Crohn's disease Plaque psoriasis Psoriatic arthritis Rheumatoid arthritis
Cosentyx	secukinumab	Ankylosing spondylitis or non-radiographic axial spondyloarthritis Enthesitis-related arthritis Hidradenitis suppurativa Plaque psoriasis Psoriatic arthritis
Enbrel	etanercept	Ankylosing spondylitis Juvenile idiopathic arthritis Plaque psoriasis Psoriatic arthritis Rheumatoid arthritis
Entyvio	vedolizumab	Crohn's disease Ulcerative colitis
Humira	adalimumab	Ankylosing spondylitis Crohn's disease

(for Humira biosimilars, see CPB 0655 - Adalimumab)		Hidradenitis suppurativa Juvenile idiopathic arthritis Plaque psoriasis Psoriatic arthritis Rheumatoid arthritis Ulcerative colitis Uveitis
Ilaris	canakinumab	Adult-onset Still's disease Gout flares Periodic fever syndromes Systemic juvenile idiopathic arthritis
Ilumya	tildrakizumab-asmn	Plaque psoriasis
Kevzara	sarilumab	Juvenile idiopathic arthritis Polymyalgia rheumatica Rheumatoid arthritis
Kineret	anakinra	Cryopyrin-associated periodic syndromes Deficiency of interleukin-1 receptor antagonist (DIRA) Rheumatoid arthritis
Olumiant	baricitinib	Alopecia areata COVID-19 in hospitalized adults Rheumatoid arthritis
Orencia	abatacept	Acute graft versus host disease Juvenile idiopathic arthritis Psoriatic arthritis Rheumatoid arthritis
Otezla	apremilast	Oral ulcers associated with Behcet's Disease Plaque psoriasis Psoriatic arthritis
Remicade (for Remicade biosimilars, see CPB 0341 - Infliximab)	infliximab	Ankylosing spondylitis Crohn's disease Plaque psoriasis Psoriatic arthritis Rheumatoid arthritis Ulcerative colitis
Rinvoq	upadacitinib	Ankylosing spondylitis or non-radiographic axial spondyloarthritis Atopic dermatitis Crohn's disease Juvenile idiopathic arthritis Psoriatic arthritis Rheumatoid arthritis Ulcerative colitis
Rituxan (for Rituxan biosimilars, see CPB 0314 - Rituximab)	rituximab	Chronic lymphocytic leukemia Granulomatosis with polyangiitis Microscopic polyangiitis Pemphigus vulgaris Rheumatoid arthritis Various subtypes of non-Hodgkin's lymphoma
Siliq	brodalumab	Plaque psoriasis
Simponi	golimumab	Ankylosing spondylitis Psoriatic arthritis Rheumatoid arthritis Ulcerative colitis
Simponi Aria	golimumab intravenous	Ankylosing spondylitis Juvenile idiopathic arthritis Psoriatic arthritis Rheumatoid arthritis

Skyrizi	risankizumab-rzaa	Crohn's disease Plaque psoriasis Psoriatic arthritis Ulcerative colitis
Stelara and biosimilars: Imuldosa, Otulfi, Pyzchiva, Selarsdi, Steqeyma, Wezlana, and Yesintek	ustekinumab ustekinumab-srlf ustekinumab-aaaz ustekinumab-ttwe ustekinumab-aekn ustekinumab-stba ustekinumab-auub ustekinumab-kfce	Crohn's disease Plaque psoriasis Psoriatic arthritis Ulcerative colitis
Taltz	ixekinumab	Ankylosing spondylitis or non-radiographic axial spondyloarthritis Plaque psoriasis Psoriatic arthritis
Tremfya	guselkumab	Plaque psoriasis Psoriatic arthritis Ulcerative colitis
Tysabri	natalizumab	Crohn's disease Multiple sclerosis
Xeljanz	tofacitinib	Ankylosing Spondylitis Polyarticular Course Juvenile Idiopathic Arthritis Psoriatic arthritis Rheumatoid arthritis Ulcerative Colitis
Xeljanz XR	tofacitinib, extended release	Ankylosing Spondylitis Polyarticular Course Juvenile Idiopathic Arthritis Psoriatic arthritis Rheumatoid arthritis Ulcerative Colitis

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